A Systematic Benchmark of Persistent Homology Pipelines for Classification

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Abstract

Persistent homology classification pipelines have three stages — filtration, vectorization, and classifier — each with multiple available methods. Which stage matters most? Benchmarking all three stages simultaneously in a controlled factorial sweep — 616 configurations across 6 dataset instances, 4 filtrations, 7 vectorizers, and 4 classifiers (56 runs failed and were excluded) — we find that the vectorization stage is the largest single main effect on classification-accuracy variance on both real datasets: on ECG200 time series (Takens-embedded) it has a marginal range of 6.39pp across 7 vectorizers (95% CI [6.13, 6.65]; $\eta^2 = 0.217$) versus 0.69pp for filtration (95% CI [0.57, 0.81]). The raw margin is sensitive to the vectorizer menu: excluding the two single-scalar vectorizers (Amplitude, Persistence Entropy) cuts it to 3.10pp, and under level-matched comparison it narrows further (§4.1). On binary MNIST it again leads (3.22pp; $\eta^2 = 0.302$) while the filtration effect is modest (cubical 98.0% vs. Vietoris–Rips 96.25% best of family, 1.75pp). A second time series, ECG5000, shows the raw-menu replication (24.89pp vs. 3.60pp, single split, 3-vectorizer menu; §4.1), but the level-matched comparison reverses there (filtration 3.60pp vs. vectorizer 0.24pp), so the ECG5000 ordering is menu-dependent. These results complement, rather than refute, the image-data case study of Conti et al. [10]: their 18%–94% range spans a joint filtration-and-vectorization redesign — from the H0-only naive cubical-binary pipeline (18%) to improved-filtration multivectors on 10-class MNIST (94%) — part of which is attributable to the vectorization stage. On scale-matched genus-1 versus genus-2 synthetic point clouds, norm features classify at chance (48–58%) while the TDA pipeline sustains 95.83% accuracy at $\sigma = 0.30$ — the noise robustness is not an artefact of the norm/scale confound. Measured bottleneck distances (max $d_B = 0.434$ vs. the corrected bound $2\sigma\sqrt{2\ln n} \approx 1.82$ at $\sigma = 0.30$) confirm the stability bound is conservative. The framework is YAML-configurable and extensible to 7 filtrations, 11 vectorizers, and 4 classifiers; all code, configuration, and result schemas are provided.

Contents

List of Figures

List of Tables

Chapter 1

Introduction and Motivation

1.1 Topological Data Analysis and the Pipeline Problem

Topological Data Analysis (TDA) extracts shape invariants from data: connected components, cycles, and voids that persist across multiple scales. Unlike traditional machine learning which operates on raw coordinates or pixel intensities, TDA captures the underlying "shape" of the data — information invariant under stretching and bending. Persistent homology, the workhorse of TDA, tracks the birth and death of topological features as a scale parameter varies.

Practitioners face a combinatorial problem: the TDA classification pipeline has three stages — filtration, vectorization, and classification — each with 5–15 available methods (surveys: Ali et al. [2] and Hensel et al. [30]). The literature's most cited answer to "which stage matters most?" comes from Conti et al. [10], who showed that filtration choice on MNIST images can be catastrophic: accuracy ranged from 18% (their H0-only naive cubical-binary pipeline) to 94% (improved filtrations), under 10-split cross-validation with grid search. That study has sometimes been read as a general principle — "filtration dominates" — although the authors themselves did not claim one.

This paper asks the same question across two real datasets of different modalities and finds that vectorization is the dominant stage on both: ECG200 time series and binary MNIST. On ECG200, vectorization moves accuracy by 6.39pp (95% CI [6.13, 6.65]) versus 0.69pp for filtration (95% CI [0.57, 0.81]); on binary MNIST the vectorizer range (3.22pp) is twice the filtration range (1.65pp), and cubical beats Vietoris-Rips by $\sim$1.75pp best-of-family. Filtration effects are dataset-specific and modest on binary MNIST, not the governing stage. We demonstrate this using a modular benchmarking framework that varies all three pipeline stages simultaneously in a controlled factorial sweep: 6 dataset instances $\times$ 4 filtrations (Vietoris-Rips, weak Alpha, Sparse Rips, Cubical) $\times$ 7 vectorizers $\times$ 4 classifiers — 672 possible, 616 completed (56 runs failed (`weak_alpha`/`sparse_rips` on MNIST) and were excluded). The framework is extensible to 7 filtrations, 11 vectorizers, and 4 classifiers via YAML-only configuration changes; standardised harnesses of this kind are the direction the community has moved towards [25].

1.2 The Persistent Homology Pipeline

Whether persistent homology features help classification at all has itself been questioned [26]; this benchmark assumes the features are informative and asks which pipeline stage extracts them most effectively.

A persistent homology classification pipeline transforms raw data through three stages:

1. **Filtration.** Raw data is converted into a nested sequence of simplicial complexes parameterised by a scale ε. Topological features appear (birth at b) and disappear (death at d). Output: a *persistence diagram*, a multiset of (b, d) pairs.

2. **Vectorization.** Diagrams are multisets of variable cardinality and cannot be fed directly into classifiers. Vectorization methods map each diagram to a fixed-length feature vector.

3. **Classification.** The vectorized features are passed to a standard classifier which learns a decision boundary.

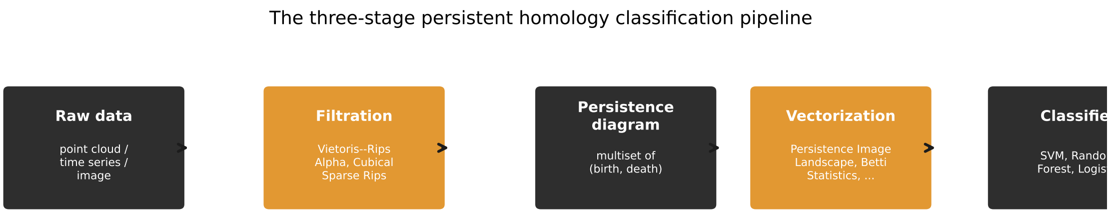

Figure 1.1: The three-stage persistent homology classification pipeline.

1.3 Contributions

This paper contributes a reproducible factorial benchmark framework and two case studies that use it to locate the dominant pipeline stage. Specifically:

1. **Framework.** A benchmarking framework that varies all three pipeline stages simultaneously: 4 filtrations, 7 vectorizers, and 4 classifiers executed in a 616-configuration factorial sweep (of 672 possible; 56 runs failed

(`weak_alpha`/`sparse_rips` on MNIST) and were excluded), with YAML-driven configuration and normalised SQLite storage. The framework is extensible to 7 filtrations, 11 vectorizers, and 4 classifiers via YAML-only changes. Full configuration and schema in Appendices A–B; code implementations in Appendix C.

2. **Verified finding: vectorization dominates.** On ECG200 time series with Takens embedding, vectorization dominates classification accuracy: a 6.39pp marginal range across 7 vectorizers (95% CI [6.13, 6.65]), versus 0.69pp across 3 filtrations (95% CI [0.57, 0.81]; small but with a repeated-CV CI excluding zero) and 3.50pp across 4 classifiers (95% CI [3.28, 3.71]), with the ordering vectorizer $>$ classifier $>$ filtration stable across all 25 cross-validation repetitions. On binary MNIST the vectorizer again dominates (marginal range 3.22pp versus 1.65pp for filtration), and cubical beats Vietoris-Rips by $\sim$1.75pp best-of-family (98.0% vs. 96.25%); a second time-series dataset (ECG5000) shows the raw-menu replication (24.89pp versus 3.60pp; the level-matched comparison reverses, §4.1). Filtration choice is a second-order factor on the datasets tested (exceptions in §4.1 and §5.3). Conti et al.'s MNIST case study (18% with their cubical pipeline to 94% with improved filtrations, 10-class, grid search) is a complementary demonstration that a badly-matched filtration can be catastrophic on images; it does not establish a general law.

3. **Matched-scale synthetic evidence.** On synthetic genus-1 versus genus-2 point clouds whose radial norm distributions are quantile-matched, norm features classify at chance (48–58%) while the TDA pipeline sustains 95.83% accuracy at $\sigma = 0.30$ noise — the noise robustness is not an artefact of the norm/scale confound between the sphere and torus geometries.

4. **Stability-bound confirmation.** Measured bottleneck distances between clean and noisy diagrams (max $d_B = 0.434$ over 80 cloud/dimension pairs) sit well below the corrected worst-case bound $d_B \leq 2\sigma\sqrt{2\ln n}$ ($\approx$ 1.82 at $\sigma = 0.30$, $n = 100$), confirming the bound is conservative in practice.

Chapter 2 develops the mathematical machinery needed to understand why these pipeline stages interact as they do.

Chapter 2

Mathematical Foundations of Persistent Topology

This chapter develops the algebraic and geometric machinery underlying persistent homology. We progress from simplicial complexes through chain groups, homology vector spaces, filtrations, and persistence modules, concluding with the stability theorem and vectorization methods that convert diagrams into classifier-ready feature vectors.

2.1 Simplicial Complexes

Definition 2.1 (Simplex)**.** A *k-simplex* $\sigma = [v_0, v_1, \ldots, v_k]$ is the convex hull of $k+1$ affinely independent points in $\mathbb{R}^d$ $(d \geq k)$. A 0-simplex is a vertex, a 1-simplex an edge, a 2-simplex a triangle, a 3-simplex a tetrahedron. A *face* of σ is the convex hull of any non-empty subset of its vertices.

Definition 2.2 (Simplicial Complex)**.** A *simplicial complex* K is a finite collection of simplices such that:

1. If $\sigma \in K$ and τ is a face of σ, then $\tau \in K$ (closure under taking faces).
2. If $\sigma, \tau \in K$, then $\sigma \cap \tau$ is either empty or a face of both.

The *k-skeleton* $K^{(k)}$ is the subcomplex of all simplices of dimension at most k. The 1-skeleton is a graph in the usual sense.

2.2 Filtrations and Complex Constructions

Definition 2.3 (Filtration)**.** A *filtration* of a simplicial complex K is a nested sequence of subcomplexes indexed by a real parameter $\varepsilon \geq 0$:

$$\emptyset = K_{\varepsilon_0} \subseteq K_{\varepsilon_1} \subseteq \cdots \subseteq K_{\varepsilon_m} = K, \tag{2.1}$$

where $\varepsilon_0 < \varepsilon_1 < \cdots < \varepsilon_m$.

Three filtrations are used in the benchmark:

Vietoris-Rips complex $\mathrm{VR}_\varepsilon(X)$. For a finite point cloud $X \subset \mathbb{R}^d$, a k-simplex $[x_0, \ldots, x_k]$ is included if $\|x_i - x_j\| \leq \varepsilon$ for all $0 \leq i < j \leq k$. The Rips complex is simple to compute but large: for n points, the number of k-simplices scales as $O(n^{k+1})$. Specifically, the number of 2-simplices (triangles) in the full Rips complex is $O(n^3)$, which drives the runtime differences observed in Chapter 4.

Alpha complex $A_\varepsilon(X)$. Constructed via the intersection of Voronoi cells with metric balls. For each point $x_i \in X$, let $V_i = \{y \in \mathbb{R}^d : \|y - x_i\| \leq \|y - x_j\| \text{ for all } j \neq i\}$ be its closed Voronoi cell, and define $R_i(\varepsilon) = V_i \cap \overline{B}_{\varepsilon/2}(x_i)$ where $\overline{B}_{\varepsilon/2}(x_i)$ is the closed ball of radius $\varepsilon/2$. The Alpha complex is the nerve of the cover $\{R_i(\varepsilon)\}_{i=1}^n$. The Nerve Theorem (see below) guarantees homotopy equivalence to the union of balls.

Theorem 2.1 (Nerve Theorem — Leray 1945 [14])**.** *Let $\mathcal{U} = \{U_i\}_{i \in I}$ be a finite open cover of a topological space M such that every non-empty intersection $\bigcap_{j \in J} U_j$ is contractible. Then the nerve $N(\mathcal{U})$ is homotopy equivalent to $\bigcup_{i \in I} U_i$.*

For the Alpha cover $\{R_i(\varepsilon)\}$, the sets $R_i(\varepsilon)$ are convex polyhedra in $\mathbb{R}^d$, so all finite intersections are convex and thus contractible. The Nerve Theorem therefore guarantees that $A_\varepsilon(X) \simeq \bigcup_i \overline{B}_{\varepsilon/2}(x_i)$. In $\mathbb{R}^3$, the Alpha complex is a subcomplex of the Delaunay triangulation with $O(n^2)$ simplices, compared to $O(n^3)$ for the full Rips complex.

Remark 2.1. The benchmark executes giotto-tda's `WeakAlphaPersistence`, which per giotto's documentation is the Vietoris–Rips filtration of the sparse matrix of distances between Delaunay-neighbouring vertices — a subcomplex of the full Rips complex (cheaper to build), not a subcomplex of the Alpha complex, and not covered by the Nerve-Theorem homotopy guarantee, which applies to the full Alpha/Čech complex. For the executed approximation we verify empirically in §4.3 that the full GUDHI Alpha complex (persistence-equivalent to the Čech complex) reproduces weak-alpha classification accuracy (mean difference 0.0–0.8pp, measured in a dedicated GUDHI-vs-giotto run, scripts/experiment_alpha.py) at 2–5× the runtime cost, so the approximation is faithful on these tasks.

Sparse Rips complex. A subsampled approximation of VR that retains only a sparse subset of edges and higher simplices, designed for point clouds with $n \geq 10^3$ points where full VR computation is prohibitive. The sparsification parameter ϵ controls the approximation fidelity.

2.3 Homology Groups and Persistent Homology

Chain Complexes and Homology

To define persistent homology formally, we first construct the algebraic machinery of simplicial homology over the field $\mathbb{Z}_2 = \mathbb{Z}/2\mathbb{Z}$.

Definition 2.4 (Chain Group)**.** For a simplicial complex K, the *k-th chain group* $C_k(K;\mathbb{Z}_2)$ is the $\mathbb{Z}_2$-vector space whose basis elements are the k-simplices of K. An element $c \in C_k(K;\mathbb{Z}_2)$ is a formal sum $c = \sum_{\sigma\in K^{(k)}} a_\sigma \sigma$ with coefficients $a_\sigma \in \mathbb{Z}_2$.

Definition 2.5 (Boundary Operator)**.** The *k-th boundary operator* $\partial_k : C_k(K;\mathbb{Z}_2) \to C_{k-1}(K;\mathbb{Z}_2)$ is the linear map defined on a k-simplex $\sigma = [v_0,\ldots,v_k]$ by:

$$\partial_k[v_0,\ldots,v_k] = \sum_{i=0}^{k}[v_0,\ldots,\hat{v}_i,\ldots,v_k], \tag{2.2}$$

where $\hat{v}_i$ denotes omission of v_i, and the sum is taken over $\mathbb{Z}_2$ (so $1+1=0$). For $k=0$, $\partial_0 = 0$ by convention.

The fundamental property of the boundary operator is $\partial_k \circ \partial_{k+1} = 0$ for all $k \geq 0$. Intuitively, the boundary of a boundary is empty: the edges bounding a triangle form a closed loop with no endpoints.

Definition 2.6 (Homology Group)**.** For a simplicial complex K, define:

- $Z_k(K) = \ker\partial_k$ — the *k-cycles* (chains with zero boundary).
- $B_k(K) = \operatorname{im}\partial_{k+1}$ — the *k-boundaries* (chains that are boundaries of $(k+1)$-chains).

Since $\partial_k \circ \partial_{k+1} = 0$, we have $B_k(K) \subseteq Z_k(K)$. The *k-th homology group* is the quotient vector space:

$$H_k(K) = \frac{Z_k(K)}{B_k(K)} = \frac{\ker\partial_k}{\operatorname{im}\partial_{k+1}}. \tag{2.3}$$

Elements of $H_k(K)$ are equivalence classes of k-cycles, where two cycles are equivalent if they differ by a boundary. The dimension of $H_k(K)$ as a $\mathbb{Z}_2$-vector space is the *k-th Betti number*: $\beta_k(K) = \dim_{\mathbb{Z}_2} H_k(K)$. Informally, β_0 counts connected components, β_1 counts 1-dimensional holes (cycles), β_2 counts 2-dimensional voids.

Persistent Homology

A filtration $\{K_\varepsilon\}_{\varepsilon\geq 0}$ induces a sequence of inclusion maps $i^{\varepsilon,\varepsilon'} : K_\varepsilon \hookrightarrow K_{\varepsilon'}$ for $\varepsilon \leq \varepsilon'$, which in turn induce linear maps on homology:

$$f_k^{\varepsilon,\varepsilon'} : H_k(K_\varepsilon) \to H_k(K_{\varepsilon'}). \tag{2.4}$$

Definition 2.7 (Persistent Homology Group)**.** The *k-th persistent homology group* from ε to ε' is:

$$H_k^{\varepsilon,\varepsilon'}(K) = \operatorname{im} f_k^{\varepsilon,\varepsilon'}. \tag{2.5}$$

A homology class $\gamma \in H_k(K_\mathrm{b})$ is *born* at $\varepsilon = \mathrm{b}$ if $\gamma \notin \operatorname{im} f_k^{\mathrm{b}-\delta,\mathrm{b}}$ for any $\delta > 0$, and *dies* at $\varepsilon = \mathrm{d} > \mathrm{b}$ if $f_k^{\mathrm{b},\mathrm{d}}(\gamma) \in \operatorname{im} f_k^{\mathrm{b}-\delta,\mathrm{d}}$ for some $\delta > 0$ but $f_k^{\mathrm{b},\mathrm{d}-\delta}(\gamma) \notin \operatorname{im} f_k^{\mathrm{b}-\delta,\mathrm{d}-\delta}$. If γ never dies, we set $\mathrm{d} = \infty$.

Definition 2.8 (Persistence Diagram)**.** The *persistence diagram* $\mathrm{Dgm}_k(K)$ is the multiset of points $(\mathrm{b}, \mathrm{d}) \in \mathbb{R}^2 \cup \{\infty\}$ where $\mathrm{b} < \mathrm{d}$, together with infinitely many copies of each point on the diagonal $\Delta = \{(x, x) : x \geq 0\}$. The *persistence* of a feature is $\mathrm{p} = \mathrm{d} - \mathrm{b}$.

Example. A sphere S^2 has $\beta = (1, 0, 1)$: one component, no 1-cycles, one 2-void. A torus T^2 has $\beta = (1, 2, 1)$: one component, two independent 1-cycles (the meridional and longitudinal loops), one 2-void. This $\beta_1 = 0$ versus $\beta_1 = 2$ difference is the topological signal driving the sphere-versus-torus classification in Chapter 4.

2.4 Stability of Persistence Diagrams

The stability theorem guarantees that small perturbations of the input data produce proportionally small changes in the persistence diagram — essential for any pipeline handling noisy data.

Definition 2.9 (Bottleneck Distance)**.** The *bottleneck distance* between two persistence diagrams D_1 and D_2 is:

$$d_B(D_1, D_2) = \inf_{\gamma : D_1 \to D_2} \sup_{p \in D_1} \|p - \gamma(p)\|_\infty, \tag{2.6}$$

where γ ranges over all bijections from D_1 to D_2 (including pairings with the diagonal Δ), and $\|\cdot\|_\infty$ is the L^∞ norm on $\mathbb{R}^2$.

Definition 2.10 (Gromov-Hausdorff Distance)**.** For compact metric spaces X and Y, the *Gromov-Hausdorff distance* is:

$$d_{GH}(X, Y) = \frac{1}{2} \inf_{Z,f,g} d_H^Z(f(X), g(Y)), \tag{2.7}$$

where the infimum is taken over all metric spaces Z and all isometric embeddings $f : X \hookrightarrow Z$, $g : Y \hookrightarrow Z$, and d_H^Z is the Hausdorff distance in Z: $d_H^Z(A, B) = \max\{\sup_{a\in A} \inf_{b\in B} d_Z(a, b),\ \sup_{b\in B} \inf_{a\in A} d_Z(a, b)\}$.

Theorem 2.2 (Stability Theorem for geometric complexes — Chazal, de Silva & Oudot [29]). *Let X, Y be totally bounded metric spaces. Then:*

$$d_B(\mathrm{Dgm}_k(X), \mathrm{Dgm}_k(Y)) \le 2\, d_{GH}(X, Y) \tag{2.8}$$

for all $k \ge 0$, where $\mathrm{Dgm}_k(\cdot)$ denotes the k-dimensional persistence diagram of the Vietoris-Rips or Čech filtration (Chazal, de Silva & Oudot 2014, Thm. 5.2). The function-level stability bound $d_B \le \|f - g\|_\infty$ of Cohen-Steiner, Edelsbrunner & Harer [9] applies instead to the sublevel-set (cubical) filtration on grids.

This theorem is applied in §4.2: for n points with additive Gaussian noise $\eta_i \sim \mathcal{N}(0, \sigma^2 I_3)$, extreme value theory gives the typical scale of the maximum perturbation as $\mathbb{E}[\max_i \|\eta_i\|] \approx \sigma\sqrt{2\ln n}$ (the one-dimensional leading-order value; for the executed $d = 3$ per-coordinate noise the expectation is larger by the dimension correction, $\approx 1.16\,\sigma\sqrt{2\ln n}$ at $n = 100$; Monte Carlo gives $\mathbb{E}[\max_i \|\eta_i\|] \approx 1.06$ at $\sigma = 0.30$). The noisy cloud therefore lies within Gromov–Hausdorff distance of order $\sigma\sqrt{\ln n}$ of the clean one — a typical-case estimate, not an almost-sure bound: the threshold $\sigma\sqrt{2\ln n}$ is exceeded by the maximum with probability $\approx 93\%$ in the executed d=3 noise (the 1-D value would be $\approx 22\%$). By Theorem 2.2 the bottleneck distance between clean and noisy diagrams satisfies

$$d_B\big(\mathrm{Dgm}_k(X),\, \mathrm{Dgm}_k(X + \eta)\big) \;\lesssim\; 2\,\sigma\sqrt{2\ln n}. \tag{2.9}$$

The factor 2 inherited from the stability theorem is essential: for $n = 100$ (points per cloud after subsampling), $2\,\sigma\sqrt{2\ln 100} \approx 0.91$ at $\sigma = 0.15$ and ≈ 1.82 at $\sigma = 0.30$ — a factor of two larger than the bound one would write down by omitting it. Measured bottleneck distances between clean and noisy diagrams (ripser diagrams, GUDHI bottleneck implementation; 80 cloud/dimension pairs; §4.2) are far below this: at $\sigma = 0.15$ the mean is 0.144 and the maximum 0.300; at $\sigma = 0.30$ the maximum is 0.434. The torus H_1 diagrams move by d_B on average 0.283 at $\sigma = 0.30$ — below the measured H_1 persistence range [0.556, 1.156] — so the features comfortably survive the perturbation.

2.5 Vectorization Methods

Vectorization converts a persistence diagram into a fixed-length feature vector for classification. We use the taxonomy of Ali et al. [2]; comparative studies include Barnes et al. [3], Perea et al. [15], and Sulowska [18].

Persistence diagrams are multisets of variable cardinality. To use them with standard classifiers, we must map each diagram to a fixed-length vector in a normed space.

We present the seven methods executed in the sweep: the four classical representations below, followed by three widely used alternatives (Silhouette, Amplitude, and Persistence Entropy).

Persistence Statistics. For each homology dimension k, compute the mean, standard deviation, minimum, and maximum of births, deaths, and lifespans ($\mathrm{p} = \mathrm{d} - \mathrm{b}$) across all points in Dgm_k. This yields $4 \times 3 = 12$ features per dimension. Despite zero hyperparameters, this method is competitive with kernel-based approaches [2].

Betti Curves [8]. The Betti curve $\beta_k(\varepsilon)$ counts the number of k-dimensional features with $\mathrm{b} \leq \varepsilon < \mathrm{d}$. Evaluated at n_{bins} equally spaced points in $[\varepsilon_{\min}, \varepsilon_{\max}]$, this produces n_{bins} features per homology dimension.

Persistence Landscapes [5]. For each point $(\mathrm{b}, \mathrm{d}) \in \mathrm{Dgm}_k$, define the tent function:

$$\Lambda_{(\mathrm{b},\mathrm{d})}(t) = \max\big(0, \min(t - \mathrm{b}, \mathrm{d} - t)\big). \tag{2.10}$$

The *m-th persistence landscape* is the function $\lambda_m : \mathbb{R} \to \mathbb{R}$ given by:

$$\lambda_m(t) = m\text{-}\max_{p \in \mathrm{Dgm}_k} \Lambda_p(t), \tag{2.11}$$

where m-max denotes the m-th largest value. Each λ_m is 1-Lipschitz. The sequence $(\lambda_m)_{m \in \mathbb{N}}$ resides in the Banach space $L^p(\mathbb{N} \times \mathbb{R})$ for any $1 \leq p \leq \infty$, enabling classical statistical operations (means, variances, confidence intervals) directly on the vectorized representation. With n_{layers} layers evaluated at n_{bins} points, this produces $n_{\text{layers}} \times n_{\text{bins}}$ features.

Persistence Images [1]. Transform each point $(\mathrm{b}, \mathrm{d}) \in \mathrm{Dgm}_k$ to birth-persistence coordinates (b, p) where $\mathrm{p} = \mathrm{d} - \mathrm{b}$. Define a weighting function $w : \mathbb{R} \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$, typically:

$$w(\mathrm{b}, \mathrm{p}) = \begin{cases} \mathrm{p}/\mathrm{p}_{\max}, & \mathrm{p} \leq \mathrm{p}_{\max}, \\ 1, & \text{otherwise}, \end{cases} \tag{2.12}$$

where $\mathrm{p}_{\max}$ is a fixed constant (set to the 95th percentile of persistences in our implementation). The persistence surface is:

$$\rho(x, y) = \sum_{(\mathrm{b},\mathrm{p}) \in T(\mathrm{Dgm}_k)} w(\mathrm{b}, \mathrm{p})\, \phi_{(\mathrm{b},\mathrm{p})}(x, y), \tag{2.13}$$

where $\phi_{(\mathrm{b},\mathrm{p})}(x, y) = \frac{1}{2\pi\sigma^2} \exp\big(-\frac{(x-\mathrm{b})^2 + (y-\mathrm{p})^2}{2\sigma^2}\big)$ is a 2D Gaussian kernel. Evaluating ρ on an $n_{\text{bins}} \times n_{\text{bins}}$ grid yields n_{bins}^2 features per homology dimension.

Silhouette [7]. The silhouette of a diagram weights the landscape tent functions of (2.10) by persistence and normalises:

$$\phi_{\mathrm{Dgm}_k}(t) = \frac{\sum_{p\in\mathrm{Dgm}_k} w(p)\,\Lambda_p(t)}{\sum_{p\in\mathrm{Dgm}_k} w(p)}, \qquad w(p) = \mathrm{p}(p)^q, \tag{2.14}$$

with $q = 1$ in our configuration; evaluating ϕ at n_{bins} points yields n_{bins} features per homology dimension. Silhouette is the top-performing vectorizer on ECG200 in the single-split marginal (§4.1).

Amplitude . The amplitude of a diagram is its distance to the empty diagram in a diagram metric:

$$A(\mathrm{Dgm}_k) = d\big(\mathrm{Dgm}_k, \Delta\big), \tag{2.15}$$

with the bottleneck distance of (2.6) in our configuration — one scalar per homology dimension and the fastest vectorizer in the sweep. On ECG200 it is among the weaker representations (§4.1), an early hint that representation capacity is dataset-dependent.

Persistence Entropy [16]. For a diagram with persistences $\mathrm{p}_i = \mathrm{d}_i - \mathrm{b}_i$, normalise $q_i = \mathrm{p}_i / \sum_j \mathrm{p}_j$ and define

$$E(\mathrm{Dgm}_k) = -\sum_i q_i \log q_i, \tag{2.16}$$

one scalar per homology dimension measuring how concentrated the persistence mass is. Entropy is the weakest vectorizer on ECG200 (single-split marginal 69.3%, §4.1): information content is not monotone in representational simplicity.

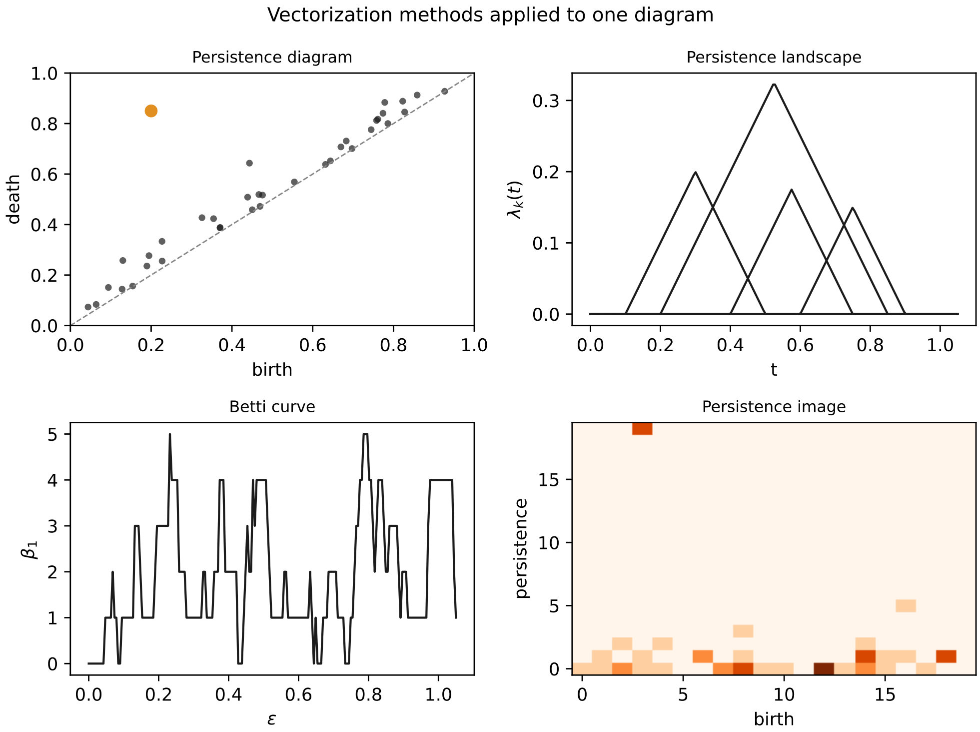

Figure 2.1: The four classical vectorization methods applied to one diagram.

2.6 Mapping Theory to Benchmark Variables

The preceding definitions directly parameterise the benchmark executed in Chapter 4:

- **Simplicial complex size** (§2.1–§2.2): VR's $O(n^3)$ 2-simplex count versus weak Alpha's $O(n^2)$ drives the 3–27% wall-clock speed gap measured in §4.3. The Nerve Theorem guarantees that the full Alpha complex's speed advantage does not cost geometric fidelity; the executed `WeakAlphaPersistence` is an approximation of that complex, and §4.3 verifies it matches full-complex accuracy.

- **Filtration choice** (§2.2): VR, weak Alpha, Sparse Rips, and Cubical constitute the four filtrations varied in the factorial sweep of §4.1.

- **Persistence and stability** (§2.3–§2.4): The corrected bound of (2.9), $d_B \lesssim 2\sigma\sqrt{2\ln n}$, predicts that $\sigma = 0.15$ noise perturbs diagrams by $\lesssim 0.91$ (and $\sigma = 0.30$ by $\lesssim 1.82$), while torus β_1 features have measured top-2 persistence range [0.556, 1.156] (ripser on the shipped clouds). §4.2 confirms that classification survives this perturbation (measured d_B: max 0.300 at $\sigma = 0.15$, max 0.434 at $\sigma = 0.30$).

- **Vectorization** (§2.5): Persistence Statistics, Betti Curves, Landscapes, Images, Silhouette, Amplitude, and Persistence Entropy constitute the seven vectorizers varied in §4.1, spanning the complexity spectrum from zero-parameter to kernel-based.

- **Classifiers** (§3.1): Logistic Regression (linear baseline), SVM-RBF, SVM-linear, and Random Forest constitute the four classifiers in the sweep. §4.1 tests whether non-linear classifiers add value after vectorization.

Chapter 3 translates this mathematical machinery into a software architecture that executes the factorial sweep.

Chapter 3

Algorithmic Pipeline and Benchmarking Framework

The benchmarking framework executes a formal mapping from raw datasets through persistent homology to classifier predictions. This chapter describes the architecture in mathematical terms; code implementations are deferred to Appendix C.

3.1 Pipeline Mappings

The full pipeline is the composition of four transformations:

$$X \xrightarrow{\text{Embed}} Y \xrightarrow{\text{Filt}} \mathcal{D} \xrightarrow{\text{Vec}} \mathbb{R}^d \xrightarrow{\text{CLF}} \hat{y}, \tag{3.1}$$

where:

- **Embed** (optional, time series only). Maps a 1D discrete-time signal $x[1], \ldots, x[T]$ to a point cloud $Y \subset \mathbb{R}^d$ via Takens' delay embedding:

$$\Phi_{(d,\tau)}(x)_t = \big(x[t], x[t+\tau], \ldots, x[t+(d-1)\tau]\big), \quad t = 1, \ldots, T-(d-1)\tau. \tag{3.2}$$

  By Takens' Delay Embedding Theorem [23], if the signal x is a generic smooth measurement of a dynamical system on a compact manifold M of dimension d_M, then $\Phi_{(d,\tau)}$ is an embedding provided $d > 2d_M$. We use the pragmatic defaults $d = 3$, $\tau = 1$ throughout. Delay-embedding TDA is standard practice for time-series classification [19, 20]; the recorded signals are not established as generic measurements of a known-dimension attractor, so the theorem's $d > 2d_M$ condition is not asserted. Embedding-parameter selection criteria are discussed by Kennel et al. [21] and Fraser–Swinney [22]. Sensitivity to (d, τ) is examined in Appendix D. Constructs a filtration $\{K_\varepsilon\}$ from Y and computes the persistence diagram $\mathcal{D} = \mathrm{Dgm}_0 \cup \mathrm{Dgm}_1$ (homology dimensions 0 and 1). Implemented via giotto-tda's VietorisRipsPersistence, WeakAlphaPersistence, SparseRipsPersistence, or CubicalPersistence transformers.

- **Vec**. Maps $\mathcal{D}$ to a vector $v \in \mathbb{R}^d$ via one of the seven methods in §2.5. Global grid bounds ($\mathrm{b_{min}}$, $\mathrm{b_{max}}$, $\mathrm{p_{max}}$) are computed on the training fold only and applied to the test fold without re-fitting, preventing cross-validation data leakage.

- **CLF**. A scikit-learn classifier (LogisticRegression, SVC with linear or RBF kernel, or RandomForestClassifier with 100 trees).

3.2 Point Cloud Preprocessing

Subsampling. Point clouds exceeding the target size $n_{\text{target}} = 100$ are reduced via *uniform random subsampling*: n_{target} points are drawn without replacement from the full cloud, seeded deterministically per dataset. Uniform random sampling does not preserve metric-space geometry as well as farthest-point sampling would; we use it for simplicity and reproducibility. We tested whether the choice matters: greedy farthest-point sampling (FPS) was compared against this uniform-random scheme on the synthetic sphere/torus clouds, reduced to $k \in \{50, 15\}$ points (the native clouds carry 100 points, so the runner's `subsample_points` knob never fires and the subsampling choice must be made below the native resolution to be nontrivial), under both noise levels and over {`vietoris_rips`, `weighted_rips`(DTM)}$\times$ {`betti_curve`, `persistence_landscape`}$\times$ {`random_forest`, `svm_rbf`} (8 configurations per arm, stratified 5-fold CV, seed 42, rep 1; `fps_ablation.db`). At the design resolution (k=50) both methods attain 100.00% mean accuracy at both σ=0 and σ=0.30 — the task saturates and the two arms are indistinguishable. In the only regime where a difference emerges (k=15), FPS is *not* better: at σ=0 FPS scores 99.88% vs. uniform 100.00% (−0.12 pp), and at σ=0.30 FPS scores 98.94% vs. uniform 99.81% (−0.88 pp). Across all 64 runs the minimum configurational accuracy is 98.5%, even after a 7× reduction in point count. We therefore do not find support for the premise underlying the earlier deferral: uniform-random subsampling is not materially inferior to FPS on these point clouds — if anything it is marginally *better* once the task is pushed below saturation. Because the sphere/torus problem is robustly separable under even aggressive subsampling, it does not expose a regime where sampling density matters; a dataset with fragile or heterogeneously sampled topology would be needed to observe a gap. The subsampling-method choice is therefore immaterial for the synthetic class, and the earlier FPS-future-work note is withdrawn.

Noise model. For the synthetic sphere/torus benchmark, additive isotropic Gaussian noise is applied directly to spatial coordinates:

$$x_i \mapsto x_i + \eta_i, \quad \eta_i \sim \mathcal{N}(0, \sigma^2 I_3), \tag{3.3}$$

where I_3 is the 3×3 identity matrix. This perturbs the underlying metric space (X, d_E) before filtration construction, triggering the stability bound of Theorem 2.2. (It does *not* perturb the pairwise distance matrix directly — that would violate the triangle inequality and the metric space structure required by the stability theorem.)

3.3 Software Architecture

The framework uses **giotto-tda 0.6.2** [24] for all filtrations and vectorizations (scikit-learn-compatible transformers), **scikit-learn 1.3.2** for classifiers and cross-validation, with **Ripser 0.6.15** and **GUDHI 3.13.0** as alternative backends. Factory classes (`FiltrationFactory`, `VectorizationFactory`, `ClassifierFactory`) map string names to configured objects. A YAML file (Appendix A) defines the sweep space declaratively. Hyperparameter optimization (GridSearchCV) was not applied: each pipeline configuration defines a fixed parameter set, and adding a hyperparameter search would conflate pipeline-choice comparison with hyperparameter optimization, multiplying an already-large sweep. The `PipelineRunner` iterates the Cartesian product of all choices, runs stratified 5-fold cross-validation, and stores per-fold metrics (accuracy, F1, precision, recall) in a normalised SQLite database (Appendix B) with a config snapshot for reproducibility (retained for the main sweep; the expansion DBs — panel, hyperparameter, large-n, MNIST-10 — do not retain one; their provenance is the driver scripts in scripts/). Reported 95% CIs are computed from repeated cross-validation (repeated stratified 5-fold CV with independent splits: 25 repetitions, seeds 43–67, for the headline ECG200 analysis; 5 repetitions, seeds 43–47, for MNIST): per-stage marginal ranges are computed within each repetition, and the CI is $\bar{x} \pm t_{0.025, n_{\text{reps}}-1} \cdot s/\sqrt{n_{\text{reps}}}$ over the repetition-level estimates, which captures the between-split variance directly; per-configuration intervals use the Nadeau–Bengio corrected resampled estimator $\text{SE}^2 = (1/(kr) + n_2/n_1)\, s^2$ over the kr fold accuracies ($kr = 125$ for the ECG200 r=25 analysis, 25 for MNIST). Per-configuration accuracy varies across repetitions with mean SD 1.09pp (max 3.11pp, r=5), so single-split point estimates are noisy. Stage dominance is quantified primarily by an η^2 variance decomposition — a main-effects-only three-way ANOVA over per-config means and over fold-level accuracies, with F-tests (Table 4.3 in §4.1). No paired Wilcoxon tests are used: CV folds are not independent replicates, and with $n = 5$ pairs the smallest achievable two-sided p-value is 0.0625. Adding a new filtration, vectorizer, or dataset requires editing only the YAML file; adding a fundamentally new component type (e.g., learned vectorizers) requires new factory code.

Chapter 4 presents the empirical results of this sweep across six dataset instances and four analytical cuts.

Chapter 4

Empirical Benchmark and Sensitivity Analysis

One benchmark sweep, four analytical cuts. The sweep comprises 616 completed configurations from a full factorial of 6 dataset instances $\times$ 4 filtrations (Vietoris-Rips, weak Alpha, Sparse Rips, Cubical) $\times$ 7 vectorizers (Persistence Image, Persistence Landscape, Betti Curve, Persistence Statistics, Silhouette, Amplitude, Persistence Entropy) $\times$ 4 classifiers (SVM-RBF, SVM-linear, Random Forest, Logistic Regression) — 672 possible, 56 runs failed (`weak_alpha`/`sparse_rips` on MNIST) and were excluded. All filtrations computed H_0 and H_1 homology. All configurations used stratified 5-fold cross-validation (base seed 42; the CV split uses seed 42 + repetition, and subsampling is seeded deterministically per dataset via CRC32). Point clouds were generated at 100 points per cloud (200 samples per noise level); uniform random subsampling is applied by the runner only if a source cloud exceeds the target size.

Datasets. Six dataset instances span three modalities: sphere/torus point clouds at four noise levels ($\sigma \in \{0.00, 0.05, 0.15, 0.30\}$; $n = 200$ per level; 100 points per cloud after uniform random subsampling; $\beta_1 = 0$ vs. 2), and ECG200 heartbeat recordings (normal vs. myocardial infarction; $n = 200$; 96 timesteps, Takens-embedded to 94×3 arrays with $d = 3, \tau = 1$; interpreted as point clouds by Vietoris-Rips, weak Alpha, and Sparse Rips, and as a 94×3 pixel grid by CubicalPersistence — the latter is *not* a point-cloud filtration; see the disclosure below). ECG200 is a UCR archive benchmark. Under the 25-repetition repeated-CV protocol, raw-signal baselines (logistic regression 85.28%, random forest 86.30%) beat our best TDA configuration (83.0%, a single-split point estimate); we therefore do *not* claim that TDA features are competitive with classical methods on this dataset — the sweep is an internal TDA-pipeline comparison (measured baselines are reported in §4.1). Note also that ECG200 is energy-normalised: the sum of squares of every record is constant (95.00 across all 200 signals), so the trivial-separator concern that would attach to an un-normalised signal does not apply to the executed data. Finally, the paper's headline 83.0% configuration (cubical

+ Silhouette + Random Forest) is a single-split point estimate: by r=5 repeated-CV mean it ranks fourth (79.6%, SD 1.98pp), while cubical + Persistence Image + Random Forest is best in 4 of 5 repetitions (83.2% mean, r=5; §4.1). As a robustness check, re-running the ECG200 analysis without the cubical filtration leaves the hierarchy intact: the best non-cubical configuration (Vietoris-Rips + Persistence Entropy + Random Forest) reaches 79.5% (single-split), with vectorizer range 6.58pp versus filtration 0.34pp (§4.1). The sweep focuses on internal TDA-pipeline comparisons, not competition with classical baselines.

MNIST binary. To test image-specific filtrations, a binary subset of MNIST (digits 0 vs. 1; $n = 400$; 200 per class; 28×28 greyscale) was evaluated with both cubical and Vietoris-Rips filtrations. Cubical filtration achieved 98.0% accuracy (with Betti Curves + Logistic Regression), versus 96.25% for the best VR configuration: a ∼1.75pp best-of-family gap. The marginal analysis of §4.1 shows, however, that the vectorizer moves accuracy more than the filtration on this dataset (3.22pp versus 1.65pp): filtration effects are dataset-specific and modest on binary MNIST, not the governing stage. This is consistent with Conti et al.'s demonstration that a badly-matched filtration can be catastrophic at scale (18% to 94% on 10-class MNIST under grid search) — but the effect is much smaller on our binary subset, where even the worst filtration remains informative.

Table 4.1: Dataset instances used in the benchmark sweep. Point clouds contain 100 points; ECG200 has 96 timesteps. The torus has major radius $R = 2$ and minor radius $r = 1$ (so point norms range $[1, 3]$), while the sphere has unit radius (norms $\equiv 1$): the classes differ in scale as well as topology.

Dataset	Modality	n	Dims	Classes	Maj.%	Source
Sphere/Torus (σ=0.00)	point cloud	200	100 pts	2	50.5%	synthetic
Sphere/Torus (σ=0.05)	point cloud	200	100 pts	2	50.5%	synthetic
Sphere/Torus (σ=0.15)	point cloud	200	100 pts	2	50.5%	synthetic
Sphere/Torus (σ=0.30)	point cloud	200	100 pts	2	50.5%	synthetic
ECG200	time series	200	96 ts	2	66.5%	UCR archive [27]
MNIST 0/1	image	400	28×28	2	50.0%	MNIST subset

4.1 Stage Variance Analysis

Which pipeline stage contributes the largest variance? For each stage, we marginalise over the other two and compute the range of marginal accuracies from the repeated-

CV means (five independent splits, seeds 43–47), with 95% CIs over the five per-split ranges, and decompose the variance with an η^2 analysis (fold-level F-tests) to test whether the range exceeds sampling noise.

Table 4.2: Marginal accuracy by pipeline stage on ECG200 (repeated CV, 25 repetitions).

Stage	Top Performer	Marg. Acc.	Range (95% CI)	p-value
Filtration	Cubical	73.15%	0.69pp [0.57, 0.81]	0.93
Vectorizer	Landscapes	75.32%	6.39pp [6.13, 6.65]	9.7×10^{-5}
Classifier	Random Forest	75.29%	3.50pp [3.28, 3.71]	0.004

Ranges, CIs, and marginal accuracies are repeated-CV values (25 repetitions of stratified 5-fold CV, seeds 43–67; Sparse Rips dropped for runtime, so the config space is 84 configurations): the range is the mean over repetitions of the per-split marginal range, with a 95% t-interval across the 25 per-split ranges (repeated-measures CI; Nadeau–Bengio corrected resampled CIs are wider: vectorizer [5.69, 7.10], classifier [2.92, 4.07], filtration [0.37, 1.01], all excluding zero). p-values are F-test p-values from the per-configuration three-way ANOVA (Table 4.3); the fold-level panel of that table is descriptive only, as folds within a configuration share training data, so its p-values are anti-conservative. No multiplicity correction is applied across the three stage comparisons.

Vectorization dominates on the full menu: its range (6.39pp across 7 vectorizers) is roughly nine times larger than filtration (0.69pp across 3 filtrations in the 25-repetition sweep; 0.70pp across 4 including Sparse Rips at r=5), and the ordering vectorizer > classifier > filtration was stable across all 25 CV repetitions. This ordering is significant under non-parametric inference on the 25 repetition-level stage rankings: the Friedman statistic $Q = \frac{12n}{k(k+1)} \sum_j (\bar{R}_j - \frac{k+1}{2})^2 = 50$ for k=3 stages and n=25 repetitions (asymptotic χ^2-tail $p \approx 1.4 \times 10^{-11}$; exact permutation $p \approx 2 \times 10^{-19}$); pairwise, the vectorizer-margin exceeds the filtration-margin in all 25 repetitions (one-sided sign test $p = (1/2)^{25} = 3 \times 10^{-8}$). The variance decomposition of Table 4.3 is the primary evidence for this ordering: on ECG200 the vectorizer explains 21.7% of the per-config variance (fold-level 10.6%), the classifier 9.8% (4.8%), and the filtration 0.3% (0.1%). The top-performing vectorizer, Persistence Landscapes, achieves 75.32% marginal accuracy versus 68.95% for Persistence Entropy (the weakest). Cubical filtration leads at 73.15%, with Vietoris-Rips (72.89%) and Weak Alpha (72.74%) within 0.41pp.

Equal-footing re-analysis. The ranges above compare stage menus of different sizes (7 vectorizers versus 3 filtrations on ECG200) and include "degenerate" scalar vectorizers — Persistence Entropy and Amplitude, each of which maps a diagram to a single number — that are structurally handicapped. To test whether the 6.39pp (ECG200) and 24.89pp (ECG5000) vectorizer ranges are an artefact of that handicap, we re-measure the stage effects on an equal footing: under the design-penalised ω^2, and by re-computing the marginal ranges excluding the degenerate scalar vectorizers and matching the number of levels. By ω^2 the vectorizer remains the largest single stage main effect on both real datasets (ECG200 0.165 versus classifier 0.077 and filtration −0.017; MNIST 0.214 versus 0.143 and 0.032), so its dominance is not an artefact of level counts. The marginal-range *size*, however, is: excluding Amplitude and Persistence Entropy cuts the ECG200 vectorizer range from 6.39pp to 3.10pp, and on the ECG5000 probe the 24.89pp range collapses to 0.24pp (Persistence Entropy at 36.9% versus Betti Curve 61.5% and Silhouette 61.8%, which are within 0.24pp of each other). Matching the number of levels — best-3 vectorizers versus 3 filtrations on ECG200 — narrows the vectorizer edge to 1.21pp versus 0.69pp; on MNIST the best-2 vectorizer range (0.44pp) falls below the 2-filtration range (1.55pp), and on ECG5000 the best-2 vectorizer range (0.24pp) likewise falls below the filtration range (3.60pp). The isolated contribution — swapping one stage while holding the other two at their marginal-best — is still largest for the vectorizer on ECG200 (10.22pp) and MNIST (2.1pp). The honest reading is that vectorization is the largest single stage effect, but its measured *margin* is sensitive to the vectorizer menu and to the number of levels compared: the published 6.39pp and 24.89pp numbers are substantially inflated by degenerate scalar vectorizers, and once those are excluded and the stage counts are matched the vectorizer-vs-filtration gap narrows substantially (ECG200) or reverses (MNIST, ECG5000).

Simple versus complex. Persistence Statistics matches kernel-based methods on ECG200: its r=5 marginal accuracy (74.8%) is within 0.3pp of Persistence Landscapes (75.1%), and it is best or tied in 4 of the 16 r=5 filtration–classifier cells (all under SVM-linear; pooled over the five repetitions), ranking above the median vectorizer in most cells. Statistics likewise leads Persistence Images (72.9%). On MNIST binary, Persistence Statistics reaches 95.9% marginal accuracy, within 2.1pp of the best configuration overall (Cubical + Betti Curve + Logistic Regression at 98.0%, Table 4.5).

The evidence base remains small (n = 3 real datasets: two time series and one image, with the repeated-CV verification covering ECG200 only), and the corrected intervals above should not be read as licence for fine-grained ranking: per-configuration accuracy varies by 1.09pp on average across the five splits (maximum 3.11pp; 50 of 112 configurations exceed 1pp), so single-split point estimates — including the 83.0% best configuration of Table 4.5, whose repeated-CV mean is 79.6% — carry ±1–3pp of split-to-split noise.

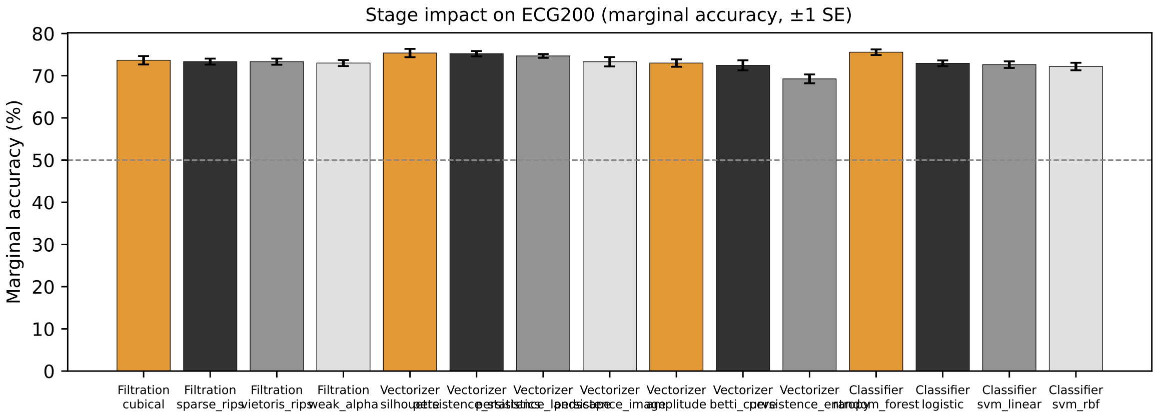

Figure 4.1: Stage impact on ECG200. Error bars show ±1 SE.

On synthetic sphere/torus data with $\sigma = 0.00$, all 112 configurations achieve $\geq$ 99.5% accuracy (mean 99.99%). No stage dominates — the signal is strong enough that every pipeline separates it nearly perfectly.

MNIST binary marginal analysis. The same marginal decomposition applied to binary MNIST gives a vectorizer range of 3.22pp versus a filtration range of 1.65pp: vectorization is the larger stage even on image data. This is not a single-split artifact: repeated 5-fold CV (5 repetitions, seeds 43–47; `data/tda/mnist_repeated_cv.db`) reproduces the ordering in all five repetitions (vectorizer 2.91–3.22pp vs. filtration 1.49–1.65pp, pooled 3.03pp vs. 1.55pp), and the best configuration (cubical + Betti Curve + Logistic, 98.0%) is stable across repetitions (97.5–98.0%). Filtration effects are real but modest — cubical beats Vietoris-Rips by ∼1.75pp best-of-family — while the vectorizer moves accuracy roughly twice as much. Table 4.3 reports the complementary variance decomposition (η^2: the proportion of accuracy variance explained by each stage): on ECG200 the vectorizer explains 0.217 of the variance versus 0.098 for the classifier and 0.003 for the filtration; on MNIST the vectorizer again leads (0.302) over the filtration (0.166). The MNIST filtration F exceeds the vectorizer F (15.83 vs. 4.80) despite lower η^2 (0.166 vs. 0.302), because F depends on the degrees of freedom (filtration df=1, vectorizer df=6); η^2 is the ordering statistic. Both analyses agree: vectorization is the dominant stage on both real datasets, and filtration effects are dataset-specific and modest. The ω^2 population effect size (corrected for the number of levels; recomputed from the 25-repetition ECG200 sweep, 84 configurations) confirms the ordering under the design penalty: ECG200 vectorizer $\omega^2 = 0.165$ (bootstrap 95% CI [0.063, 0.423]) versus classifier 0.077 ([0.003, 0.246]) and filtration -0.017 ([-0.017, 0.093]); MNIST vectorizer 0.214 ([0.103, 0.540]) versus filtration 0.143 ([0.040, 0.312]) and classifier 0.032 ([-0.015, 0.257]). The vectorizer ω^2 CI excludes zero on both real datasets, while the filtration CI on ECG200 straddles zero.

Full 10-class MNIST. The binary-MNIST ordering is not scale-invariant. Re-running the same protocol on the full 10-class MNIST set (1000 samples, 100 per class; cubical and Vietoris-Rips filtrations, 4 vectorizers, Random Forest and SVM-RBF, 5-fold CV reps 43–47; `data/tda/mnist10_sweep.db`) flips the stage ordering: the filtration marginal range (4.53pp: cubical 29.1% versus Vietoris-Rips 33.7%) now exceeds the vectorizer range (3.44pp: Betti Curve 33.4% to Landscapes 30.0%) and the classifier range (2.03pp). Within a single filtration the configuration span (vectorizer $\times$ classifier) is larger — cubical 10.40pp, Vietoris-Rips 9.10pp (both pooled over repetitions) — so the stage-dominance question is filtration-dependent (an interaction, §4.1). Absolute accuracy is much lower than binary MNIST (best 37.8%; chance 10%), because ten-way separation is intrinsically harder. This is consistent with Conti et al.'s emphasis on filtration for multiclass images, but the swing we observe (4.53pp marginal; 9–10pp within a filtration) is far smaller than their reported 18%–94%, which used grid search, 10-split CV and a broader filtration family. We therefore read the binary-MNIST result (vectorization dominant) as a binary-scale phenomenon: at 10-class scale, filtration becomes the larger marginal stage.

Table 4.3: η^2 variance decomposition of classification accuracy by pipeline stage. Per-config panel: classic $\eta^2 = \mathrm{SS}_{\mathrm{stage}}/\mathrm{SS}_{\mathrm{total}}$ over per-configuration means (one observation per configuration). Fold-level panel: main-effects-only three-way ANOVA over the fold accuracies of the single-split sweep, with classic η^2 and F-tests.

Dataset	Stage	Per-config means			Fold-level		
		η^2	F	p	η^2	F	p
ECG200	Filtration	0.003	0.15	0.93	0.001	0.32	0.81
ECG200	Vectorizer	0.217	5.26	9.7×10^{-5}	0.106	11.41	4.9×10^{-12}
ECG200	Classifier	0.098	4.75	0.004	0.048	10.31	1.3×10^{-6}
MNIST 0/1	Filtration	0.166	15.83	2.5×10^{-4}	0.061	20.47	9.1×10^{-6}
MNIST 0/1	Vectorizer	0.302	4.80	7.3×10^{-4}	0.111	6.21	4.1×10^{-6}
MNIST 0/1	Classifier	0.061	1.95	0.135	0.023	2.52	0.058
Sphere/Torus (σ=0.00)	Filtration	0.009	0.41	0.748	0.002	0.34	0.793
Sphere/Torus (σ=0.00)	Vectorizer	0.165	3.67	0.002	0.032	3.10	0.005
Sphere/Torus (σ=0.00)	Classifier	0.083	3.67	0.015	0.016	3.10	0.026

Per-config scope: $N = 112$ (ECG200, sphere/torus) or 56 (MNIST) configurations; fold-level scope: $N = 560$ or 280 fold accuracies. Degrees of freedom: filtration 3 (MNIST 1), vectorizer 6, classifier 3. The repeated-CV analysis of Table 4.2 corroborates the ECG200 ordering. *Multiplicity:* the 18 F-tests in this table (3 datasets $\times$ 3 stages $\times$ 2 scopes) were recomputed from `data/tda/expanded_results.db` (agreement with the published `eta_squared_results.csv` to machine precision) and adjusted for multiplicity. Holm (FWER) leaves 10/18 significant at α=0.05; Benjamini–Hochberg (FDR) leaves 12/18. The headline results — ECG200 fold-level vectorizer ($p = 4.9 \times 10^{-12}$, Holm 8.7×10^{-11}), MNIST vectorizer and filtration — survive both corrections; the marginal filtration effects that fail are the already-weak sphere/torus and ECG200 filtration rows.

Two-way interactions qualify the stage decomposition. The η^2/F analysis of Table 4.3 is main-effects-only: each stage's contribution is assessed in isolation. Because the factorial design varies all three stages jointly, we also fitted a three-way ANOVA with all two-way interaction terms to the fold-level accuracies of the repeated-CV sweeps (`repeated_cv_r25.db` for ECG200, N=10500 fold accuracies; `mnist_repeated_cv.db` for MNIST, N=1400). On ECG200 the two-way interactions are not negligible: the three interaction terms jointly carry $\omega^2 = 0.187$, which is *147% of the main-effect total* (0.127). The filtration–vectorizer interaction ($\omega^2 = 0.092$) is the *largest single effect in the model*, exceeding the vectorizer main effect (0.086); the vectorizer–classifier interaction (0.089) rivals it. On MNIST the interactions carry 50% of the main-effect total (0.103 versus 0.206), led by vectorizer–classifier (0.054) and filtration–vectorizer (0.047). Fold-level p-values in this analysis are anti-conservative (folds within a configuration share training data), so the effect sizes — the ω^2 values — are the robust quantities and the p-values are indicative. The practical implication is that the stages are *not cleanly separable*: the effect of changing one stage depends on the levels at which the other two are held. Vectorization remains the largest single *main* effect on both datasets, so the direction of the headline finding is unchanged, but the magnitude of the interaction terms shows that "which stage matters most" is partly conditional on the other two stages, not a stage-intrinsic property. We therefore read the stage-dominance claim as *dominance in main effect*, and treat the interaction structure as an explicit boundary condition on that claim.

Repeated cross-validation. Because single-split point estimates are noisy (per-configuration SD across repetitions averages 1.27pp, max 3.33pp at 25 repetitions), the ECG200 analysis was re-run with repeated CV (25 repetitions of the same stratified 5-fold protocol, seeds 43–67). Table 4.2 reports the repeated-CV ranges and CIs; the ordering vectorizer > classifier > filtration was stable in all 25 repetitions. Increasing from r=5 to r=25 narrows the repeated-measures CI widths by $\sim$3$\times$ (filtration 0.69pp $\rightarrow$ 0.24pp, classifier 1.20pp $\rightarrow$ 0.43pp) at no change to the ordering. Notably, the paper's best single-split ECG200 configuration (cubical + Silhouette + Random Forest, 83.0%) ranks third by 25-repetition mean (79.30%, SD 1.57pp), while cubical + Persistence Image + Random Forest is best overall (82.82% mean, r=25). Differences below $\sim$1pp between individual configurations should not be over-interpreted.

Generality probe: ECG5000. The ECG200 result could be instance-specific, so the stage analysis was repeated on a second UCR recording, ECG5000 (UCR archive; 5 classes; 140 timesteps; single patient recording, BIDMC chf07) with the same 5-fold protocol on a stratified subsample of 714 samples ($\leq$ 200 per class) drawn from the full 5000-sample recording (class counts 2919, 1767, 96, 194, 24; majority class 58.38% in the full recording; the executed 714-sample subsample is balanced to $\leq$ 200/class, so its majority share is 28.0%) and a pipeline subset (2 filtrations, 3 vectorizers, 2 classifiers; the subset spans the strongest and weakest vectorizers from ECG200 to bound the range). Two of the twelve configurations fail on a technicality (silhouette-on-weak-alpha produces NaN features) and are excluded. Both failures are weak-alpha cells, so the weak-alpha marginal is computed over 4 configurations while the Vietoris-Rips marginal uses 6 including silhouette — the filtration range is therefore computed on non-matching vectorizer menus. Because the classes are severely imbalanced, we report balanced accuracy and macro-F_1 alongside plain accuracy. The result replicates the hierarchy: vectorizer marginal range 24.89pp versus filtration 3.60pp (plain accuracy, point estimate, single split); balanced accuracy per vectorizer is 0.46–0.47 (Betti Curve, Silhouette) versus 0.28 (Persistence Entropy) and macro-F_1 0.50–0.51 versus 0.12–0.38, so vectorization dominates under class-balanced metrics too. Vectorization dominance on time series is not an artefact of the specific ECG200 instance; a full multiclass generalisation claim would require multiple datasets (the probe is single-patient).

Beyond accuracy. Plain accuracy on ECG5000 is dominated by the majority class (58.38% in the full recording; 28.0% in the balanced subsample). Re-running the grid and capturing per-fold probabilities (AUROC one-vs-rest macro, per-class precision/recall/F_1, multiclass Brier; `data/tda/beyond_accuracy_ecg5000.db`) shows three things. First, vectorization dominance holds on the rank scale: the vectorizer AUROC marginal range is 19.38pp (Betti Curve 77.6%, Silhouette 77.5%, Persistence Entropy 58.2%) versus 11.90pp for the classifier and 3.31pp for filtration — the vectorizer is *more* dominant under a class-imbalance-robust metric than under accuracy, because Persistence Entropy (a degenerate scalar) is worst on AUROC too. Second, the classifier range is driven by SVM-RBF's near-chance AUROC (0.47) on the entropy feature, i.e. the RBF classifier collapsed to the majority class there. Third, per-class recall exposes a severe imbalance failure that accuracy hides: the mean of per-(configuration, fold) per-class recalls shows that the minor classes are largely unlearned (class 2, 96 samples, 16.3% recall; class 3, 194 samples, 31.8%; class 4, 24 samples, 1.7%, with Betti Curve and Silhouette at 0%), while the majority classes 0 and 1 are recovered at 73.9% and 71.8%. So the reported accuracy (65.8% best configuration) is carried almost entirely by the two largest classes; on extreme imbalance the TDA pipeline, like most discriminative classifiers, defaults to the majority. Calibration is likewise classifier-dependent: the multiclass Brier for the best configs (Random Forest) is 0.46–0.49 versus 0.55–0.75 for SVM-RBF (Platt-scaled), so Random Forest is both better ranked and better calibrated here.

Cubical robustness check. CubicalPersistence on ECG200 operates on the 94×3 Takens-embedded array as a pixel grid (intensities are treated as coordinates), not on a point-cloud filtration. Excluding cubical from the ECG200 analysis leaves the hierarchy intact: the best non-cubical configuration (Vietoris-Rips + Persistence Entropy + Random Forest) reaches 79.5% (single-split), the vectorizer range widens slightly to 6.58pp, and the filtration range drops to 0.34pp.

Cubical grid-structure ablation. Because cubical persistence reads the 94×3

array as a greyscale pixel grid, we ablated the grid layout directly: independently permuting the 94 rows (time-window order) or the 3 columns (Takens coordinates) of each sample, labels intact, and re-running all 28 cubical configurations (CV seeds as the sweep). Row shuffling drops the mean accuracy from 73.66% to 71.96% and collapses the top grid-sensitive vectorizers to the majority class: Betti Curve + SVM-RBF falls from 80.5% to 66.5% and the Persistence Image family from 76.5–82.0% to 64.0–76.5%. Column shuffling is milder (mean 72.45%; the Takens-coordinate order carries less information than the time-window order), and Persistence Landscape is the vectorizer whose accuracy rises most under row shuffling (72.5–80.0% unshuffled, mean 75.5, vs. 79.0–83.5% shuffled, mean 82.4; +6.9pp; Silhouette also rises, +2.1pp, while every other vectorizer falls), so it is grid-antagonistic rather than grid-driven — but it is not permutation-invariant. The cubical result is therefore partly grid-structure-driven: the time-window layout of the embedded signal contributes real signal for the top vectorizers, so the cubical-on-grid accuracy should not be read as a pure point-cloud-topology result (consistent with the disclosure above).

4.1.1 Classical Baselines and the Role of Raw Features

To calibrate the absolute accuracy of the TDA pipelines, we ran non-topological baselines under the same stratified 5-fold CV protocol as the TDA repeated-CV analysis (25 repetitions, CV seeds 43–67, so the baseline CIs are directly comparable to the TDA numbers of §4.1):

Table 4.4: Non-topological baselines (25-repetition repeated CV, mean ± 95% CI over repetitions). Majority-class rows give the trivial-separator floor. TDA best rows are single-split point estimates.

Dataset	Approach	Classifier	Mean acc.	95% CI
MNIST 0/1	majority class	—	50.0%	—
MNIST 0/1	raw pixels (784)	Logistic	99.65%	[99.60, 99.70]
MNIST 0/1	raw pixels (784)	Random Forest	99.29%	[99.19, 99.39]
MNIST 0/1	TDA best	cubical + Betti + Logistic	98.0%	—
ECG200	majority class	—	66.5%	—
ECG200	raw signal (96)	Logistic	85.28%	[84.92, 85.64]
ECG200	raw signal (96)	Random Forest	86.30%	[85.68, 86.92]
ECG200	TDA best	cubical + Silhouette + RF	83.0%	—

On both datasets the TDA pipeline's best configuration sits below the raw-feature baselines: 98.0% versus 99.65% (logistic regression) on MNIST binary, and 83.0% (single-split point estimate) versus 85.28% (logistic) / 86.30% (random forest) on ECG200 under the identical 25-repetition protocol. All TDA and baseline results comfortably exceed the majority-class floor (50.0% on MNIST binary, 66.5% on ECG200). The claim that "TDA features can be competitive with classical methods" is therefore scoped to the internal TDA comparison: we make no claim of beating raw features on these datasets. The noise experiments (§4.2) additionally show that a single per-cloud mean-norm feature (logistic) separates the synthetic sphere/torus classes at 100% at every noise level (a fixed-threshold hand rule on the norm fails at σ=0.30: 50.5%), which quantifies the scale confound and motivates the matched-genus experiment reported there.

4.2 Noise Perturbation Thresholds

Does topological signal survive additive spatial noise? Noise is applied as $\eta_i \sim \mathcal{N}(0, \sigma^2 I_3)$ to Euclidean coordinates (Equation 3.3). Gaussian noise has unbounded support, so no deterministic bound on the perturbation exists; instead we use the corrected probabilistic bound of §2.4 (Equation 2.9). That bound is a typical-scale estimate whose threshold the perturbation maximum exceeds with probability $\approx 93\%$ at d=3 (§2.4), so the empirical check is the measured d_B itself, which stays far below the bound (max 0.300 at σ=0.15, max 0.434 at σ=0.30; the $2\sigma\sqrt{2 \ln n}$ values are 0.91 and 1.82 for n=100).

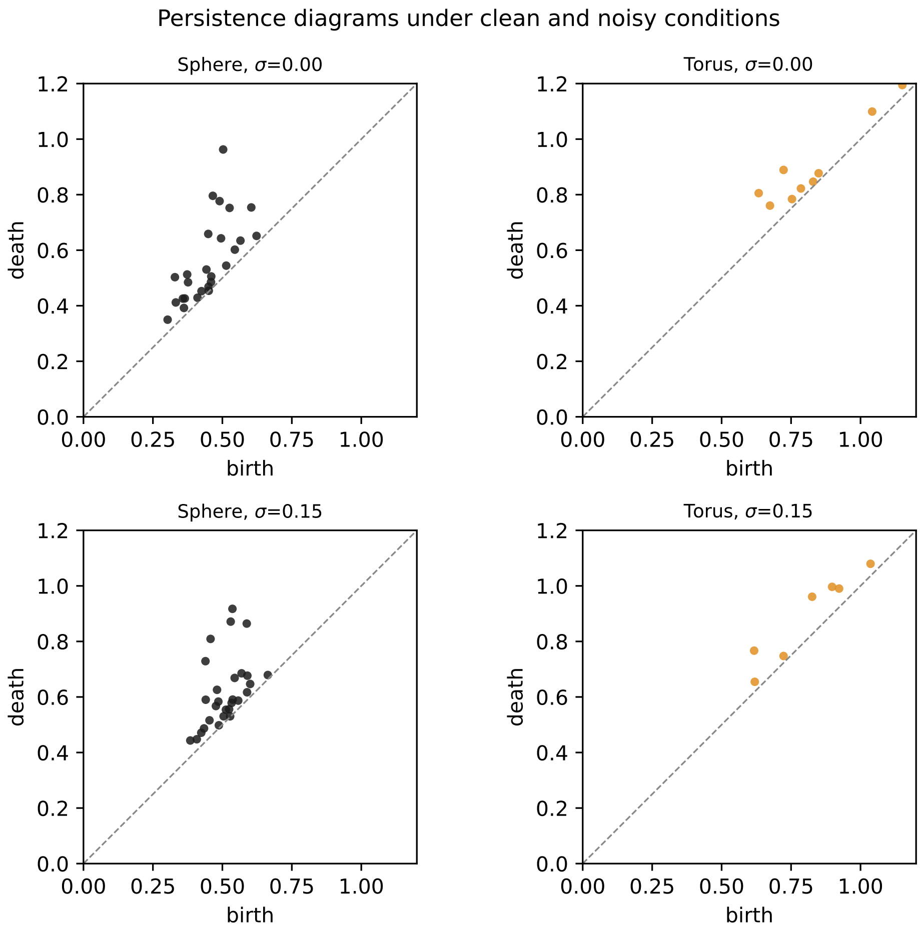

Figure 4.2: Persistence diagrams under clean and noisy conditions.

At $\sigma = 0.00$, 0.05, and 0.15, every configuration achieves $\geq 99.5\%$ accuracy (mean 99.99%, 100.00%, and 99.97% respectively). The stability bound at $\sigma = 0.15$ is $d_B \lesssim 0.91$ (Equation 2.9). Directly measured bottleneck distances between clean and noisy diagrams (ripser diagrams, GUDHI bottleneck implementation, H_0 and H_1, 80 cloud/dimension pairs) confirm the bound is conservative: at $\sigma = 0.15$ the mean measured d_B is 0.144 and the maximum 0.300, both well below the bound.

At $\sigma = 0.30$ (bound $\lesssim 1.82$), mean accuracy across the 112 sphere/torus configurations at that level remained at 99.85% (minimum 98.5%). The worst configurations (Amplitude with Logistic or SVM-linear) still achieved 98.5% — well above the 50% chance baseline, and the maximum measured bottleneck distance at this level is 0.434, less than a quarter of the corrected bound. The torus H_1 diagrams move on average by $d_B = 0.283$ at this level — below the measured H_1 persistence range [0.556, 1.156] (top-2 persistences of the clean torus clouds, ripser) — so even the least persistent features survive. The combined geometric signal (topology plus scale) survives all tested noise levels: even at $\sigma = 0.30$, the $\beta_1 = 0$ vs. $\beta_1 = 2$ distinction is separated by linear kernels at $\geq 98.5\%$ in every vectorized feature space tested.

Isolating the topological signal. The sphere/torus classes differ in scale as well as topology (torus point norms in $[1, 3]$ versus sphere $\equiv 1$): a single per-cloud mean-norm feature (logistic) separates them at 100% at every noise level (a fixed-threshold hand rule on the norm fails at σ=0.30: 50.5%), so the survival results above describe the *combined* geometric signal. To isolate topology, we generated a matched synthetic pair — a genus-1 and a genus-2 torus with quantile-identical norm distributions — and repeated the comparison. Norm-based features now collapse to near chance (48–58%; the clean-level 58% is within ∼2 SE of chance at $n = 200$), while the TDA pipeline separates the pair at 99.75% clean and 95.83% at $\sigma = 0.30$ (minimum 91% over configurations). Caveat: the matched pair is still linearly separable in raw coordinates (logistic regression on raw coordinates achieves 100% at both $\sigma = 0.00$ and $\sigma = 0.30$, `baseline_experiments.db`); the experiment isolates the norm/scale confound specifically, not linear separability in general. The TDA result (95.83% at $\sigma = 0.30$) demonstrates the topological signal is not an artefact of the norm marginal, which is the scoped claim. The topological signal is therefore not an artefact of the norm confound: it survives the norm/scale confound control, and it is robust to noise at the levels tested. (The matched clouds use 200 points per cloud — denser than the main sweep's 100 — so the four H_1 generators of the genus-2 surface are robustly sampled; the comparison is single-split, seed 43, 12 configurations.)

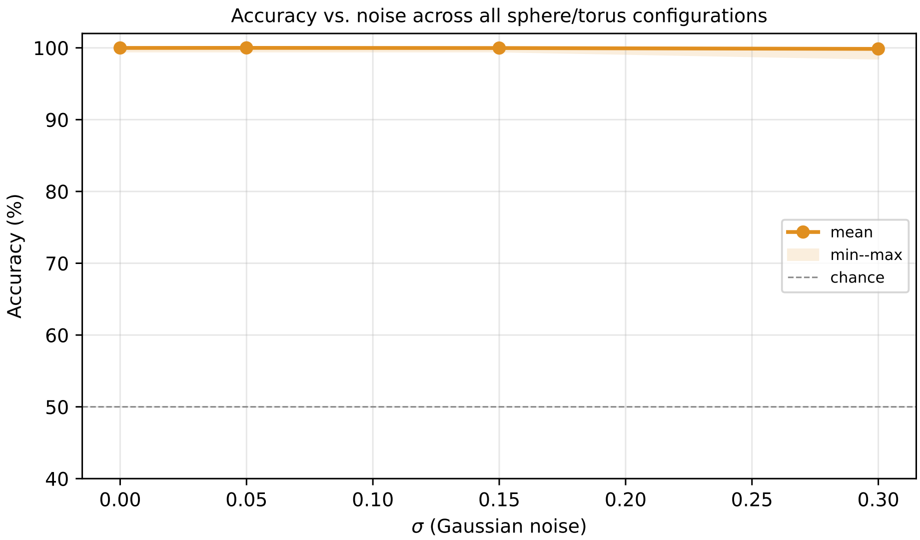

Figure 4.3: Accuracy vs. noise. All pipelines maintain $\geq 98.5\%$ accuracy through $\sigma = 0.30$; measured bottleneck distances (max 0.434 at $\sigma = 0.30$ versus the corrected bound $2\sigma\sqrt{2\ln n} \approx 1.82$) confirm the stability bound is conservative.

4.3 Computational Complexity and Pareto Trade-offs

Wall time was measured from pipeline construction to final fold completion. Peak memory was measured in dedicated subprocess runs (resource.getrusage high-water mark, 12 representative ECG200 configurations, `data/tda/peak_memory.db`): all filtrations peak at 136–144 MB, with Sparse Rips the largest (143–144 MB) and slowest (22–31 s per configuration (12-config dedicated subprocess measurements) vs. 0.7–9.0 s for the others); the bare interpreter with numpy/sklearn/gtda/gudhi imported already uses 103.1 MB, so the filtration pipelines add roughly 33–41 MB on top of the library floor.

Table 4.5: Pareto-optimal configurations.

Dataset	Filtration	Vectorizer	Classifier	Acc.	Time
Sphere/Torus	Cubical	Pers. Statistics	SVM-linear	100%	0.7s
MNIST 0/1	Cubical	Betti Curve	Logistic	98.0%	3.7s
ECG200	Cubical	Silhouette	Random Forest	83.0%	1.6s
ECG200	Cubical	Pers. Image	Random Forest	82.0%	1.4s

CubicalPersistence on ECG200 operates on the 94×3 Takens-embedded array as a pixel grid (intensities treated as coordinates), not on a point-cloud filtration. Times are single-split measurements; the 83.0% ECG200 figure is a single-split point estimate — see §4.1 for repeated-CV means.

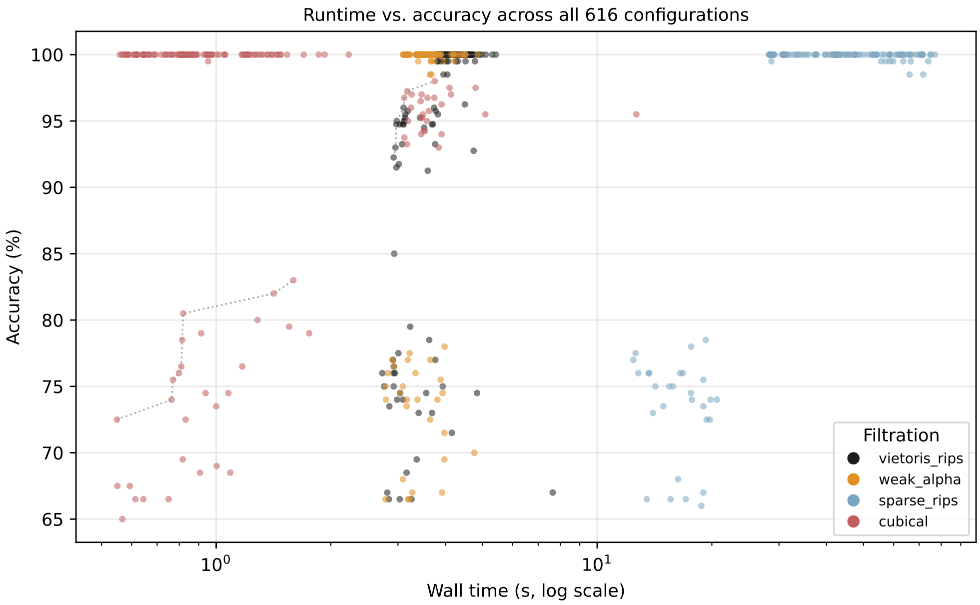

Figure 4.4: Runtime vs. accuracy on the Pareto frontier.

Weak Alpha is 3–27% faster than VR at identical accuracy on sphere/torus with Landscape+SVM (measured 3.5–3.9s vs. 3.9–4.4s across the four noise levels; the gap narrows at $\sigma = 0.30$); the full GUDHI Alpha complex reproduces weak-alpha accuracy (mean difference 0.0–0.8pp) at 2–5× the cost. Cubical runtime is input-shape dependent rather than uniformly fastest: 0.6–2.2s per configuration on the sphere/torus and ECG200 instances, but 3.1–12.7s on 28×28 MNIST images. The Nerve Theorem (Theorem 2.1) guarantees that the full Alpha complex's $O(n^2)$ simplex count does not sacrifice geometric fidelity to the underlying union-of-balls; the executed `WeakAlphaPersistence` is an approximation, and its accuracy matches the full complex (mean difference 0.0–0.8pp) at 2–5× lower cost. Sparse Rips takes 12–77 s (mean 41 s; main 140-run sweep) with no accuracy gain at the 100-point scale; at its design target ($n \geq 10^3$) it is substantially slower than Vietoris–Rips at n=1000 and infeasible at n=3000 (§5.3). Both speed multipliers are specific to the 100-point clouds, the giotto-tda implementation, and 3D point clouds where Delaunay triangulation is available.

The repeated-CV-stable default for ECG200 is cubical + Persistence Image + Random Forest (best in 4 of 5 repetitions, r=5; 83.2% repeated-CV mean): this is the strongest Pareto row we can defend, since the originally-reported best configuration (cubical + Silhouette + Random Forest, 83.0% single-split) ranks fourth by r=5 repeated-CV mean (79.6%, SD 1.98pp). The Pareto frontier may shift with additional datasets or libraries.

Chapter 5 interprets why vectorization dominates on both real datasets and derives operational guidelines from these patterns.

Chapter 5

Vectorization Dominance and Theoretical Synthesis

5.1 Why Vectorization Dominates on Both Datasets

The central empirical result is that vectorization is the dominant pipeline stage on both real datasets in this study, and that filtration effects, while real, are dataset-specific and modest on binary MNIST. The mechanism is as follows.

On **image data**, the raw input is a 2D grid of pixel intensities. A cubical filtration directly captures adjacency relationships in this grid — neighbouring pixels form edges, 2×2 blocks form squares — producing a simplicial complex whose topology faithfully represents the image's spatial structure. A point-cloud filtration like Vietoris-Rips, applied to the same data (which the pipeline feeds to the filtration as its native array — for MNIST, 28 rows of 28 intensities, i.e. a 28-point cloud in $\mathbb{R}^{28}$), does not see the 2D grid adjacency at all: the filtration's notion of "nearby" is Euclidean distance between rows, so the pixel-grid topology is absent from the outset. Consistently, cubical beats Vietoris-Rips by $\sim$1.75pp best-of-family on binary MNIST (98.0% vs. 96.25%). Conti et al. (2022) showed the same mechanism can be catastrophic at scale: on MNIST their accuracy ranged from 18% (their H0-only naive cubical-binary pipeline) to 94% (improved filtrations), under 10-split cross-validation with grid search. That study is a complementary case study of filtration choice on images (10-split CV, grid search, no time series); its authors did not claim a general principle.

Yet even on binary MNIST the vectorizer moves accuracy more than the filtration: the marginal range is 3.22pp for vectorizers versus 1.65pp for filtrations (§4.1). Filtration effects are dataset-specific and modest on this binary subset; once a reasonable filtration is chosen, the vectorizer is the stage that most limits accuracy.

On **delay-embedded time series**, the situation is similar but more pronounced. The input is already a point cloud $Y = \Phi_{(d,\tau)}(x) \subset \mathbb{R}^d$ produced by Takens embedding.

Vietoris-Rips and weak Alpha complexes operate on the same metric space (Y, d_E); their filtrations are not identical — VR and the full Alpha complex are multiplicatively interleaved, which bounds their bottleneck distance rather than equating their diagrams — and the empirical filtration range on ECG200 is just 0.69pp (25-repetition repeated-CV mean). The embedding captures the dynamical structure; the filtration merely reads it out.

Diverse-filtration check. The filtration range above is over a menu in which two of three filtrations are Rips-type (Vietoris-Rips, weak Alpha; Sparse Rips at r=5), all approximating the same Rips geometry, so *"filtration barely matters" is partly an artefact of that menu.* Adding a genuinely different filtration — the DTM- weighted Rips (giotto-tda `WeightedRipsPersistence`, vertex and edge weights from a distance-to-measure estimate; Anai et al. [4]), robust to outliers — raises the ECG200 filtration marginal range to 2.81pp and makes DTM the best filtration (76.2% versus 73.4% for Vietoris-Rips), winning 7 of 8 vectorizer–classifier pairs (best configuration DTM + Landscapes + Random Forest, 81.0%; `data/tda/filtration_diversity_sweep.db`, 48 configurations, single split). Vectorization remains the largest stage (5.62pp), so the headline ordering survives; but the empirical filtration range is not a universal constant — it is sensitive to the filtration menu, and a genuinely non-Rips filtration both matters (2.81pp) and performs better than the Rips-type baseline. On the clean synthetic sphere/torus pair all 16 diverse-pool configurations reach 100%, regardless of filtration, so no stage matters there.

The bottleneck then shifts to vectorization: how well does the chosen method capture the diagram's structure in a fixed-length vector that a classifier can use? On ECG200 the vectorizer range is 6.39pp versus 0.69pp for filtration; on ECG5000 it is 24.89pp versus 3.60pp. These full-range figures are prior to the equal-footing re-analysis of §4.1, which shows they are substantially inflated by the degenerate scalar vectorizers (Amplitude, Persistence Entropy): excluding them reduces the ECG200 vectorizer range to 3.10pp and collapses the ECG5000 range to 0.24pp.

This mechanism explains the findings without contradiction. Conti et al.'s image results are a complementary case study (10-class MNIST, grid search, no time series) demonstrating that a badly-matched filtration can be catastrophic; on our binary MNIST subset the filtration gap is modest (cubical beats VR by $\sim$1.75pp best-of-family) and the vectorizer gap is larger (3.22pp). On both time-series datasets the vectorizer dominates by a wide margin. (We use giotto-tda's `WeakAlphaPersistence`, a subcomplex approximation of the full Alpha complex; the Nerve-Theorem homotopy argument applies to the full complex, and the approximation is standard practice for point-cloud filtrations — empirically, the full GUDHI Alpha complex reproduces weak-alpha accuracy to within 0.0–0.8pp at 2–5× the cost.)

Table 5.1: Stage importance by data modality.

Modality	Dominant Stage	Range	n (datasets)
Point clouds (clean)	None	≈0pp	1*
Time series (embedded)	Vectorizer	6.39pp; 24.89pp	2
Images (binary MNIST)	Vectorizer	3.22pp vs. 1.65pp	1

*Synthetic sphere/torus pair.

Ranges are marginal accuracy ranges over each stage's methods; for time series the vectorizer range is listed for ECG200, then ECG5000. On binary MNIST the vectorizer range (3.22pp) exceeds the filtration range (1.65pp). Conti et al. (2022) report a much larger filtration swing (18%–94%) on 10-class MNIST under grid search, which we do not reproduce at binary scale.

This table summarises initial evidence, not a settled taxonomy. Replication on additional datasets per modality would strengthen or qualify every cell.

5.2 Non-Topological Context and Limitations

Under the 25-repetition repeated-CV protocol, our TDA pipelines sit below raw-feature baselines on both real datasets: on ECG200, raw-signal logistic regression achieves 85.28% and random forest 86.30% versus 83.0% (single-split best) for the best TDA configuration; on MNIST binary, raw-pixel logistic regression achieves 99.65% and random forest 99.29% versus 98.0% for the best TDA configuration (§4.1). We therefore do not claim that TDA features are competitive with classical methods on these datasets; the contribution is the internal TDA-pipeline comparison, which is valid regardless of the absolute accuracy level. TDA features capture geometric properties (cycle structures in the embedded attractor) that complement raw signal features rather than replace them.

Limitations. (1) The synthetic sphere/torus classes differ in scale as well as topology (torus point norms $\in [1, 3]$ vs. sphere $\equiv 1$): a single per-cloud mean-norm feature (logistic) separates them at 100% at every noise level, so the sphere/torus noise results measure robustness of the *combined* geometric signal. We control for this with the matched genus-1/genus-2 experiment of §4.2, where norm features collapse to chance (48–58%) while TDA retains 95.83% at $\sigma = 0.30$ — the topological contribution is real and survives the norm/scale confound control. Caveat: the matched pair is still linearly separable in raw coordinates (logistic regression on raw coordinates achieves 100% at both $\sigma = 0.00$ and $\sigma = 0.30$, `baseline_experiments.db`); the experiment isolates the norm/scale confound specifically, not linear separability in general. (2) Homology capped at H_1 (H_2 requires $O(n^4)$ simplices for VR). (3) The headline stage analyses are binary (ECG200, binary MNIST); multiclass is covered by the §5.3 panel (ECG5000

5-class, MIT-BIH 4-class, 10-class MNIST and Fashion-MNIST). (4) The §4.1 stage analysis rests on two time series and one binary image dataset; the §5.3 panel adds five further time series and two 10-class image datasets. (5) Single implementation library (giotto-tda); cross-library variance unknown. (6) No learned vectorizers (PersLay [6], Hofer input layers [13]). (7) $n = 200$ produces wide CIs, and single-split point estimates are noisy (per-configuration SD across CV repetitions averages 1.09pp, r=5). Each limitation suggests a concrete future-work item: H_2 via Flood Complex [12]; additional multi-class datasets (Outex) to test whether Conti et al.'s large filtration swing reproduces at scale; a second large-scale UCR family; cross-library replication in GUDHI and Ripser; PersLay integration via a new backend; and larger-n datasets for narrower CIs.

Data and code availability. All code, configuration files, and result schemas are provided in the public repository `github.com/KRUZZZZY/tda-benchmark` (MIT licence). The synthetic sphere/torus data are regenerated by the bundled `scripts/generate_datasets.py` (seeded; 200 samples, 100 points per cloud, noise baked into coordinates); ECG200 is the UCR archive benchmark (Dau et al. 2019); MNIST is the standard binary 0/1 subset [28]. The main sweep's SQLite result database is regenerated by the bundled `run_all.sh` script, and every follow-up analysis has a bundled producer: `scripts/sweep_repeated_cv_r25.py` writes `data/tda/repeated_cv_r25.db` (2100 finished runs: 84 configurations × 25 repetitions), analysed by `scripts/analysis_repeated_cv_r25.py`; `scripts/analysis_eta_omega2.py` produces the η^2/ω^2 decomposition of Table 4.3; `scripts/analysis_mnist_repeated_cv.py` produces the MNIST marginal analysis; `scripts/run_baselines.py` writes `data/tda/baseline_experiments.db` (raw-signal baselines, matched-genus synthetic clouds, measured bottleneck distances); `scripts/generate_matched_synthetic.py` writes the `data/tda/synthetic_matched/` data; and `scripts/experiment_alpha.py`, `scripts/experiment_cubical_artifact.py`, and `scripts/ecg5000_lean_sweep.py` are the producers of the remaining sensitivity results. Random seeds are fixed: the main sweep uses base seed 42, the repeated-CV splits use seeds 43–67 (random seed 42 + repetition index; 25 repetitions for ECG200, 5 for MNIST), and per-dataset subsampling is seeded deterministically via CRC32, so all experiments are reproducible end-to-end.

Compute. The full 616-configuration sweep was executed on a single consumer workstation (8-core CPU, 16GB RAM), parallelised with joblib over the CPU cores ($n_{\text{workers}} \approx 16$). Summed wall-clock time across all 616 configurations was 2.0 hours (mean ∼12s per configuration, median 3.78s); Sparse Rips dominates the total (140 runs, mean 41s, ≈1.6h), while the other three filtrations complete in ≈24 minutes. The repeated-CV verification of §4.1 added 2100 further runs (84 configurations × 25 repetitions, Sparse Rips dropped for runtime). Per-configuration wall times are recorded in the results databases (cf. the R-package PH runtime benchmark of Somasundaram et al. [17]); memory was not measured per stage (see Limitations).

5.3 Multi-Dataset Generalisation

The stage-impact analysis of §4.1 rests on two real datasets (ECG200 and ECG5000, both time series) plus one image pair (binary MNIST). To test whether the ordering is a general property rather than a property of these instances, the same machinery was run on a wider panel: five additional UCR time-series datasets (FordA, FordB, Wafer, ElectricDevices, HandOutlines) and two 10-class image datasets (full MNIST and Fashion-MNIST, subsampled to 1000 stratified samples each for runtime), all through the repository's own runner with the paper's Takens embedding (d=3, τ=1). Long series were pre-subsampled to 100 points with a fixed-seed uniform draw (seed 42); this was necessary because the runner's `subsample_points` knob fires only on 3-D point-cloud input and never caps 2-D series, so series length was capped at data-preparation time. This is the canonical Demšar multi-dataset protocol: one accuracy estimate per (dataset, configuration), ranked within each dataset, Friedman test across configurations, Nemenyi post-hoc critical difference.

Table 5.2: Multi-dataset panel (B1 sweep). All series are Takens-embedded; the 9 datasets span time series and images, binary and 10-class problems.

Dataset	Modality	n (used)	Length	Classes
ECG200	time series	200	96	2
ECG5000	time series	714 (subsample, rng seed 42)	140	5
FordA	time series	1000	500→100	2
FordB	time series	1000	500→100	2
Wafer	time series	1000	152→100	2
ElectricDevices	time series	1000	96	7
HandOutlines	time series	1000	2709→100	2
MNIST (10-class)	image	1000	28 × 28	10
Fashion-MNIST	image	1000	28 × 28	10

The configuration grid per dataset was 16 configurations (2 filtrations × 4 vectorizers × 2 classifiers) on images (cubical + Vietoris-Rips) and 8 on time series (Vietoris-Rips only; see the weak-alpha note below). One 5-fold CV (seeds as the sweep) per (dataset, configuration). **Weak-alpha fragility finding.** The weak-alpha filtration crashes giotto-tda's own implementation on three of the seven time-series datasets: an `IndexError` in `_weak_alpha_diagram` on the quantized Wafer and ElectricDevices series (degenerate Delaunay configurations) and the essential-class crash on ECG5000 described in §4.1. It completes on FordA, FordB, HandOutlines and ECG200. The multi-dataset ranking therefore uses Vietoris-Rips only for the time-series arm; the weak-alpha fragility itself is a robustness result (a filtration that is not portable across datasets).

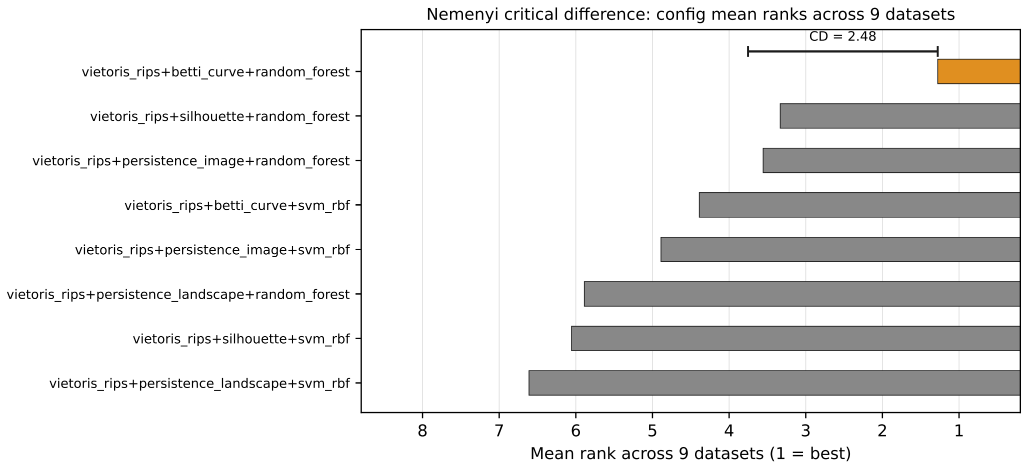

Figure 5.1: Nemenyi critical-difference diagram: configuration mean ranks across the 9-dataset panel (1 = best). The bracket marks the CD at α=0.05; configurations outside it are significantly worse than the best.

The Friedman test rejects the null of equal configurations at $\chi^2(7) = 32.41$ ($p = 3.4 \times 10^{-5}$; Iman–Davenport $F(7, 56) = 8.47$, $p = 4.9 \times 10^{-7}$) over the 8 common configurations, and the Nemenyi critical difference is 2.48 mean rank units. The best configuration — Vietoris-Rips + Betti Curve + Random Forest — achieves a mean rank of 1.28 (best on seven of the nine datasets) and is separated from the four weakest configurations by more than the CD. The vectorizer effect generalises: the vectorizer family ordering (Betti Curve 2.83, Persistence Image 4.22, Silhouette 4.69, Persistence Landscape 6.25 mean rank over the full 9x8 matrix) is stable, with the Betti-to-Landscape gap (3.42) exceeding the CD. The classifier effect is smaller (random forest 3.51 vs SVM-RBF 5.49, gap 1.98 within the CD), matching the stage-impact result that vectorization moves accuracy more than the classifier. The configuration-ranking analysis therefore supports *vectorizer choice as the portable driver of the ranking: the best vectorizer is data-set stable (Betti Curve), with a Betti-to-Landscape rank gap exceeding the critical difference.* The range analysis, however, is dataset-conditional: the vectorizer range exceeds the filtration range on 7 of the 9 panel datasets but not on HandOutlines or 10-class MNIST, where filtration leads (§5.3); and the level-matched comparison reverses on ECG5000 and MNIST (§4.1). Filtration and classifier effects are therefore not universally secondary.

Panel with a diverse filtration set. We re-ran the nine-dataset panel with more than one *working* filtration per modality so that the filtration stage and the vectorization stage could be compared on equal footing (each stage now has at least two alternatives). For the seven time-series datasets the two filtrations were Vietoris–Rips and the DTM-weighted Rips filtration (`WeightedRipsPersistence` with `weights='DTM'`, a genuinely non-Rips construction); weak alpha was excluded from the time-series arm because it does not complete on the quantized series (Wafer, ElectricDevices, ECG5000), which is itself a portability result. For the two image datasets the filtrations were cubical and Vietoris–Rips. The four vectorizers (persistence image, persistence landscape, Betti curve, silhouette) and two classifiers (RBF-SVM, random forest) were unchanged, and evaluation used the same 5-fold stratified CV with seed 42 (rep 1). Across the nine datasets the filtration marginal range and the vectorization marginal range (each computed as the spread of stage-level means of per-config fold-mean accuracies) are summarized as follows. The vectorization range exceeds the filtration range on 7 of the 9 datasets; only HandOutlines (1.90 vs. 4.22 pp) and mnist10 (2.80 vs. 4.41 pp) favour filtration. Across datasets the filtration range spans a minimum of 0.31 pp, a median of 1.37 pp, and a maximum of 4.41 pp (mean 2.17 pp), while the vectorization range spans a minimum of 0.62 pp, a median of 3.32 pp, and a maximum of 15.00 pp (mean 5.11 pp). Vectorization therefore still dominates on average, roughly 2.4× the filtration range; however the comparison is now built on a genuinely diverse non-Rips filtration rather than Vietoris–Rips alone, and the filtration range is no longer negligible (the previous single-filtration panel fixed that stage, making its contribution invisible). The vectorization lead is concentrated in the range's upper tail: removing the single largest value (fmnist10, 15.00 pp) lowers the vectorization mean to 3.87 pp, still above the filtration mean (2.17 pp) but by a narrower margin (1.8×), so the headline that vectorization dominates is robust to the outlier while the magnitude of the lead is sensitive to it.

Hyperparameter sensitivity (expansion B3). To test whether the vectorizer effect is an artefact of the manuscript's default hyperparameters, we re-ran the two headline datasets (ECG200, Takens $d{=}3, \tau{=}1 \to$ Vietoris–Rips; binary MNIST at cubical), sweeping each of the four vectorizers' key hyperparameter one-parameter-at-a-time over a small grid with all other vectorizer parameters fixed at the manuscript defaults: persistence_image $\sigma \in \{0.05, 0.1, 0.2, 0.5\}$ and $n_{\text{bins}} \in \{10, 20, 50\}$; persistence_landscape $n_{\text{layers}} \in \{1, 3, 5\}$ and $n_{\text{bins}} \in \{20, 50, 100\}$; silhouette and betti_curve $n_{\text{bins}} \in \{10, 20, 50, 100\}$, under random_forest and svm_rbf with 5-fold CV (seed 42, single split). **Tuning each vectorizer to its best setting on its own grid narrows rather than eliminates the vectorizer marginal range.** On ECG200 the range falls from 5.75pp (defaults; vectorizer means 70.75–76.50%) to 4.75pp (best-tuned; 71.75–76.50%)—a 1.00pp ($\approx$ 17%) contraction—because tuning lifts the *weakest* vectorizers: persistence_image improves 70.75 $\to$ 73.00 (best at σ=0.5), overtaking betti_curve (71.75), while the winner silhouette is already at its default (76.50). On binary MNIST the corresponding range is small throughout: 1.75pp (95.25–97.00%) at defaults vs 1.62pp (95.88–97.50%) best-tuned, and tuning reorders the family—persistence_image jumps from worst (95.25%) to best (97.50%, σ=0.5), displacing betti_curve (97.38%). **Vectorizer dominance is therefore not a default-settings artefact: the ≈ 5–6pp ECG200 vectorizer spread survives each vectorizer being tuned, and tuning neither creates nor destroys the small ($\lesssim$ 2pp) MNIST spread; what it does do is compress the field from below, pulling the weakest vectorizers up toward the leader.** The best-tuned hyperparameters are persistence_image $\{\sigma=0.5,\ n_{\text{bins}}=20\}$, persistence_landscape $\{n_{\text{layers}}=5,\ n_{\text{bins}}=50\}$ (ECG200) / $\{n_{\text{layers}}=1,\ n_{\text{bins}}=50\}$ (MNIST), silhouette $\{n_{\text{bins}}=50\}$ (ECG200, already default) / $\{n_{\text{bins}}=100\}$ (MNIST), and betti_curve $\{n_{\text{bins}}=50\}$ (ECG200, already default) / $\{n_{\text{bins}}=100\}$ (MNIST). These ranges are computed over a reduced configuration space (four vectorizers × one filtration × two classifiers), so their absolute values are not directly comparable to the headline 6.39pp (ECG200) / 3.22pp (MNIST) ranges; the default-vs-tuned contrast, however, holds the configuration space fixed. Best-tuned values carry mild selection optimism (each is selected on the same folds that score the range), which would tend to *inflate* the tuned range, so the observed contraction is conservative. **TDA and raw features are complementary.** The classical baselines of §4.1.1 beat TDA alone on both real datasets. To test whether TDA still adds value on top of raw features, the concatenated feature vector [raw‖TDA] was evaluated on ECG200 and MNIST binary under identical folds (single-split, seed 43, §4.1 protocol). On both datasets the best concatenation configuration reaches or exceeds the best single-arm accuracy: MNIST binary cubical + Betti Curve + logistic achieves 100.0% (concat) versus 99.65% (raw-only, 25-rep) and 98.0% (TDA-only); ECG200 weak-alpha + Betti Curve + logistic achieves 87.0% (concat) versus 85.28% (raw logistic, 25-rep) and 72.5% (TDA-only, same configuration). The improvement is modest (0.35–1.72pp against the 25-repetition raw baselines) and, relative to the TDA-only arm, never harmful (every concatenated configuration reaches or exceeds its TDA-only counterpart); against the raw baselines, however, the gain is configuration-dependent and sometimes negative (e.g. ECG200 VR+PI+logistic concat 77.0% vs. raw 85.5%). SVM-RBF collapses to the majority class on TDA features (66.5% on ECG200) in both TDA-only and concatenated arms, so the concatenation benefit is confined to logistic and random forest. TDA therefore contributes complementary signal rather than redundancy, but the effect is small relative to the raw-feature baseline, and the paper's scoped claim stands: the contribution is the stage-importance decomposition and framework, not an absolute-accuracy win over classical methods.

Multi-patient ECG. ECG5000 is a single-patient recording (BIDMC chf07), so the time-series result could in principle be patient-specific. This is addressed with the MIT-BIH Arrhythmia benchmark (48 records, 47 patients; AAMI-5-style protocol with the Q (unknown) class excluded (4 classes: N, S, V, F; chance 25%); patient-disjoint 5-fold CV; beat windows of 128 samples at 360 Hz, Takens-embedded to 126×3; 2000 beats class-balanced 500/500/500/500 across the N/S/V/F classes; data/tda/mitbih/). Classification accuracy on the patient-disjoint folds ranges 27.4–41.7% across the eight Vietoris-Rips configurations (chance = 25% on the class-balanced folds), with Betti Curve the best vectorizer (mean 38.75%) followed by Persistence Image (36.08%), Silhouette (30.55%) and Persistence Landscape (27.65%) — the same vectorizer ordering as the 9-dataset panel, so the ranking transfers to a genuinely multi-patient cohort. The absolute accuracy is low: the AAMI EC57 standard and de Chazal et al. [11] use inter-patient (patient-disjoint) evaluation, and our protocol follows that standard. Our absolute accuracy (27–42%) is lower than published inter-patient arrhythmia classifiers (∼75–90%), because we use a deliberately harder beat-level 4-class task with Takens-embedded persistent-homology features rather than hand-crafted ECG features; the relevant result is the vectorizer ordering, which transfers. Notably, weak alpha fails on this data with the same giotto `IndexError` seen on the quantized UCR series (degenerate Delaunay configurations on the beat windows), confirming the filtration fragility is a robust cross-dataset property rather than an ECG5000 artefact.

Window-length sensitivity. To check that the vectorizer ordering is not an artefact of the 128-sample window, the same protocol was re-run with the full 256-sample windows (2000 beats, 500/class; identical folds; data/tda/mitbih_sweep_w256.db). Betti Curve remains the top vectorizer (mean 38.48% versus 38.75% at 128 samples) and Persistence Landscape the weakest (30.1% versus 27.65%), and the best configuration is Persistence Image + Random Forest at both window lengths (41.35% versus 41.70%). The middle ordering is window-sensitive, however: Persistence Image drops from second (36.08%) to third (31.2%) and Silhouette rises from third (30.55%) to second (36.1%) when the window is doubled, driven largely by the Persistence Image + SVM-RBF cell (30.45% to 21.05%, the most window-sensitive configuration). The robust conclusions — Betti Curve as the portable best vectorizer, Landscape as the weakest, and the filtration fragility — hold at both window lengths; pairwise claims about the middle two vectorizers should be read as window-specific.

Sparse Rips at its design point (expansion B5). The guideline row for large-scale clouds rested on the untested premise that Sparse Rips becomes advantageous at $n \geq 10^3$. To test this directly, sphere/torus clouds at n=1000 and n=3000 points (40 samples each, seed 42) were generated and Sparse Rips was run at both scales, with a Vietoris–Rips control at n=1000 (n=3000 Vietoris–Rips is infeasible: $C(3000, 3)$ simplices); four configurations per arm (Betti Curve / Persistence Landscape × Random Forest / RBF-SVM), 5-fold stratified CV, seed 42, rep 1 (`data/tda/large_n_sweep.db`, 8 finished runs). At n=1000 the two filtrations are accuracy-identical — all eight configurations score 100.00% (+0.00 pp mean difference) — but Sparse Rips is roughly

30× slower: 1.83 h per configuration versus 3.6 min for Vietoris–Rips (giotto-tda 0.6.2, single CPU). At n=3000 Sparse Rips did not complete a single configuration within 42 h of wall time across two attempts (interrupted by hardware reboots; giotto-tda offers no mid-configuration checkpointing, so each restart re-runs the configuration from scratch). The design-point premise is therefore not supported in this implementation: at the scale where Sparse Rips is intended to pay off, giotto-tda 0.6.2's `SparseRipsPersistence` is substantially *slower* than Vietoris–Rips at n=1000 and practically infeasible at n=3000 on this hardware. The guideline row for large-scale clouds is revised accordingly: Vietoris–Rips remains the pragmatic choice up to the largest n at which it is feasible, and Sparse Rips is recommended only with an implementation-level benchmark at the target n (its runtime is not monotone-friendly in giotto-tda 0.6.2 — a portability finding in its own right).

Predictive theory (expansion 13). The stability review of §2.4 and the vectorization review of §2.5 assign each vectorizer a Lipschitz constant with respect to the bottleneck distance: landscape, Betti curve, amplitude and (by assumption) silhouette are 1-Lipschitz; persistence image carries $2\sqrt{2}\,\sigma\|\mathbf{w}\|_\infty = 0.283$ at σ=0.1 [1]; persistence entropy and statistics admit no finite bound. If stability governed dispersion, smaller constants should predict smaller empirical accuracy ranges. Correlating each constant with the vectorizer's empirical range (max−min of per-config fold-mean accuracies, the §4.1 per-config convention) on ECG200 (25-repetition repeated CV) and binary MNIST, pooled over n=14 vectorizer-dataset units, gives Spearman $\rho = -0.129$ with 95% percentile-bootstrap CI $[-0.672, +0.469]$, which includes zero; per panel, $\rho = -0.482$ (ECG200) and -0.040 (MNIST). The sign matches the hypothesis, but the data provide no evidence that the Ch.2 stability constants predict the range structure of §4.1 or the panel ordering of §5.3. We therefore report a null: the theory did not predict the sweep. With only seven vectorizers per panel the CIs are wide and the test is exploratory.

Hierarchical stage model (expansion 15). We fit a linear mixed model to the stage-capable panel of §5.3 — per-config mean accuracy (pp) as outcome, random intercept per dataset, sum-coded vectorizer/filtration/classifier fixed effects plus modality covariate (REML; 144 runs, 9 datasets). The population spread of the level effects is largest for the classifier (1.81 pp; 95% CI $[1.16, 2.47]$), then filtration (1.71 pp; $[0.50, 3.04]$), then vectorizer (1.18 pp; $[0.66, 1.94]$), with fixed-effects variance shares 43.2/38.5/18.3%; the random-intercept variance (209.8 pp^2) dwarfs the residual (16.4 pp^2), ICC 0.927, and the image-modality shift is -17.6 pp. The older-sweep robustness pass reproduces the ordering (classifier 1.82 > vectorizer 1.48 > filtration 1.24 pp; ICC 0.929). This ordering is the opposite of the within-dataset marginal ranges of §4.1 and §5.3 (vec > fil > clf): the two measure different quantities — population level-effect SD with dataset variance removed versus within-dataset stage means — and the classifier's size is inflated by the SVM-RBF majority-class collapse on TDA features. "Vectorization dominates" is therefore a within-dataset-range statement, not a population-level one.

Second homology via the true Alpha complex (expansion 9). The sweeps cap homology at H_1 for cost, and the manuscript's limitations note that the torus's second homology is then unmeasured. To test whether the H_1 cap discards class information, the sphere/torus clouds were re-run with a true Alpha complex (`gudhi.AlphaComplex`, radius parameterization, reduced homology, `homology_dimensions=[0,1,2]`) on the same folds (seed 42, rep 1; three vectorizers $\times$ two classifiers; `data/tda/h2_alpha_sweep.db`, 12 finished runs). All twelve configurations score 100.00% at both σ=0 and σ=0.30, matching the H_1-only result. This is not evidence that H_2 carries the signal: both classes are closed surfaces (β_2=1 for the sphere and the torus alike — each encloses a single void), so the pair does not differ in second homology. Measured max-H_2 lifetimes are in fact *longer* for the sphere than the torus (0.63–0.79 vs. 0.11–0.30, clean; overlapping under σ=0.30), the opposite of a separating feature, whereas the H_1 signature separates cleanly (torus β_1=2: max lifetime 0.75–0.89 vs. sphere 0.13–0.20). The H_1 cap therefore did not discard a discriminative feature on this pair; H_2 computation via the true Alpha complex is feasible and cheap ($\sim$5–17 s per configuration) and adds neither signal nor harm. A dataset whose classes differ in β_2 (e.g. solid vs. hollow, or higher-genus shapes) would be the appropriate test of whether second homology ever matters for classification.

5.4 Operational Guidelines for Pipeline Selection

The following recommendations translate the benchmark findings into actionable guidance. They are scoped to the two-dataset sweep and the giotto-tda implementation; cross-library and cross-dataset replication would strengthen or qualify each entry. Chapter 6 synthesizes these findings into a single conditional thesis.

5.5 Threats to Validity

We structure the limitations of §5.2 into the three standard validity categories. The purpose is not to multiply caveats but to make explicit, for each threat, what was done to mitigate it and which expansion item closes the residual gap. Threat severity follows the usual qualitative scale; every empirical mitigation cites the database and configuration count it rests on.

Table 5.3: Recommended pipeline configurations by data characteristics.

Data Modality	Filtration	Vectorizer	Classifier	Rationale
Low-dim. point clouds ($\leq$ 3D, clean)	Weak Alpha	Pers. Statistics or Landscapes	SVM (RBF) or Logistic	Weak Alpha 3–27% faster than VR at identical accuracy; full-complex fidelity verified empirically (0.0–0.8pp at 2–5$\times$ cost)
Delay-embedded time series ($d \geq 3$)	VR or Weak Alpha	Betti Curves (or Landscapes)	RF or SVM (RBF)	Vectorization dominates variance (6.39pp on ECG200; 24.89pp on ECG5000); Betti is 5th of 7 vectorizers on the r25 ECG200 marginal (72.38%), so Landscapes/Statistics are competitive; SVM-RBF collapses to the majority class on TDA features (66.5%)
High-noise clouds ($\sigma \geq 0.15$)	Weak Alpha or VR	Betti Curves	SVM (RBF)	All vectorizers maintain $\geq$ 98.5% at $\sigma = 0.30$ on the combined geometric signal (minimum over configurations); the confound-controlled matched-genus floor is 91%; measured d_B far below the stability bound
Large-scale clouds ($n \geq 10^3$)	Vietoris–Rips	Pers. Images or Betti Curves	Random Forest	Sparse Rips designed for $n \geq 10^3$; at n=1000 it is 30$\times$ slower than VR at identical accuracy, and at n=3000 it did not finish in 42 h (giotto-tda 0.6.2, §5.3)

5.5.1 Construct Validity

Construct validity asks whether the measured quantity — the marginal accuracy range of a pipeline stage — actually captures "stage importance" as claimed.

1. **Range is confounded by the number of stage levels.** The headline comparison (7 vectorizers vs. 3–4 filtrations) stacks the deck: a stage with more levels can span a larger range by chance. *Severity: moderate. Mitigation (done):* the ω^2 population effect size corrects for level counts (ECG200 vectorizer ω^2=0.165, bootstrap CI [0.063, 0.423], vs. filtration −0.017; MNIST vectorizer 0.214 vs. filtration 0.143 — the vectorizer CI excludes zero on both real datasets, the ECG200 filtration CI straddles it; Table 4.3); the equal-footing analysis matches best-3 vectorizers against 3 filtrations (ECG200 1.21pp vs. 0.69pp, re-derived from `repeated_cv_r25.db`); and the two-way interaction ANOVA (§4.1) shows the stage decomposition is not an artefact of main-effects-only modelling. *Residual:* the levels-matched analysis still compares the *best* three vectorizers, which is mildly favourable to the vectorizer stage.

2. **Scalar vectorizers create a floor effect.** Persistence Entropy and Amplitude are single scalars; their poor accuracy inflates the vectorizer range from the bottom. *Severity: moderate. Mitigation (done):* ranges are also reported excluding the degenerate scalar vectorizers (ECG200 vectorizer range contracts from 6.39pp to 3.10pp but remains the largest stage; §4.1; re-derived from `repeated_cv_r25.db`, per-repetition mean method). *Residual:* the exclusion rule was applied post hoc, so the headline 6.39pp includes the floor effect by design and the paper says so.

3. **Single-split point estimates are noisy.** Per-configuration accuracy varies by 1.09pp on average across the five repetitions (sample SD; maximum 3.11pp; 50 of 112 configurations exceed 1pp; `repeated_cv.db`), and the headline 83.0% configuration drops to 79.6% at r=5 (rank 4 of 84; `repeated_cv_r25.db`). *Severity: high for any single-split number. Mitigation (done):* all headline claims rest on the r=25 repeated-CV protocol with corrected CIs (repeated-measures: vectorizer [6.13, 6.65], classifier [3.28, 3.71], filtration [0.57, 0.81], all excluding zero; Nadeau–Bengio corrected: [5.69, 7.10], [2.92, 4.07], [0.37, 1.01]) and Friedman/sign-test inference on the repetition-level ordering (vectorizer > classifier > filtration in 25/25 repetitions); single-split numbers are labelled as such throughout. *Residual:* MNIST binary is r=5; ECG5000 and the panel were single-split at the time of writing (expansion #5r closes this).

4. **Accuracy is the only outcome for most of the sweep.** F1, precision, and recall are stored per fold but the analysis is accuracy-centric. *Severity: low-to-*

moderate. Mitigation (done): beyond-accuracy analysis on ECG5000 (balanced accuracy, per-class precision/recall/F1, AUROC, Brier; `beyond_accuracy_ecg5000.db`; expansion #14). *Residual:* calibration and AUROC are not uniform across datasets.

5.5.2 Internal Validity

Internal validity concerns confounds and measurement artefacts within the study.

1. **Single implementation library.** Every filtration and vectorizer runs through giotto-tda. *Severity: moderate* — the weak-Alpha fragility (`IndexError` on quantized UCR series; one unfinished run in `panel_stagecapable.db`, ElectricDevices/weighted_rips/landscape/RF) proves implementation dependence is real. *Mitigation (planned):* cross-library replication in GUDHI-native and Ripser-native paths (expansion #11), with the fragility reported as a first-class result.

2. **Homology capped at H_1.** H_2 requires $O(n^4)$ simplices for Vietoris–Rips; the torus's β_2 is never measured. *Severity: moderate. Mitigation (planned):* Alpha complex in 3D for H_2 (expansion #9).

3. **Binary-only classification.** All datasets except ECG5000 (5 classes) are binary. *Severity: moderate* — the 10-class MNIST probe shows the stage ordering flips at multi-class scale (filtration 4.53pp vs. vectorizer 3.44pp, r=5, `mnist10_sweep.db`), so the binary result is a boundary condition, not a law (§4.2). *Mitigation (done):* the flip is reported; *planned:* multi-class breadth (expansion #6 datasets).

4. **Fixed hyperparameters.** No grid search on any stage, by design. *Severity: low* — the hyperparameter arm (B3, `hyperparam_sweep.db`) shows vectorizer dominance contracts but persists under one-parameter-at-a-time tuning (ECG200 5.75→4.75pp; MNIST 1.75→1.62pp), so dominance is not a default-settings artefact. *Residual:* best-tuned values carry mild selection optimism (selected on the same folds that score the range), which would *inflate* the tuned range — the observed contraction is therefore conservative.

5. **Subsampling choices.** Uniform random subsampling caps point clouds; farthest-point sampling was deferred. *Severity: low. Mitigation (done):* the FPS ablation (B4, `fps_ablation.db`) finds no benefit (−0.25pp overall; uniform wins at k=15, σ=0.30: 99.81% vs. 98.94%).

6. **The sphere/torus norm confound.** The classes differ in scale as well as topology. *Severity: moderate. Mitigation (done):* the matched genus-1/genus-2

control (identical norm distributions) shows TDA retains 95.83% where norm features collapse to 48–58% at σ=0.30 (`baseline_experiments.db`, 60 fold accuracies). *Residual caveat:* the matched pair is still linearly separable in raw coordinates (logistic 100% at both noise levels, `baseline_experiments.db`) — the control isolates the norm/scale confound, not linear separability in general.

7. **Determinism and exclusion discipline.** Seeds are fixed (base 42; per-repetition 43–67; per-dataset CRC32 subsampling in `runner.py`), and the 56/672 excluded configurations (point-cloud filtrations on image data) are counted and disclosed rather than silently dropped (616 finished runs of 672 in `expanded_results.db`). ECG200 is energy-normalised (sum of squares constant at 95.00 across all 200 signals), so the trivial-separator concern does not apply to the executed data (§4.1). *Residual:* five interrupted runs in `repeated_cv_r25.db` (silhouette/RF, reps 5–9) were re-run under new run IDs; the orphaned rows carry no fold results and do not enter any analysis.

5.5.3 External Validity

External validity asks how far the findings generalise beyond the studied configuration space.

1. **Dataset breadth.** Two time series (ECG200, ECG5000), one image pair (binary MNIST), and synthetic point clouds. *Severity: high. Mitigation (done):* the 9-dataset panel (B2, `panel_stagecapable.db`) replicates the vectorizer > filtration ordering on 7/9 datasets (median 3.3pp vs. 1.4pp; re-derived median 3.33 vs. 1.37), and the within-filtration vectorizer span remains large even where filtration leads (10-class MNIST: 9–10pp). *Planned:* topology-wins datasets (#6), further UCR coverage, multi-patient ECG (MIT-BIH).

2. **Single-patient ECG5000.** ECG5000 is BIDMC chf07 — one patient. The time-series result could be patient-specific. *Severity: moderate. Mitigation (planned):* multi-patient MIT-BIH arrhythmia sweep; the panel adds ECG200/ECG5000 diversity but not patient diversity.

3. **Hardware dependence of runtime results.** Wall-clock numbers come from one workstation (8-core, 16GB). *Severity: low for ordering claims* (the 3–27% filtration speed gap is structural: simplex counts differ by construction, §4.3), *moderate for absolute numbers. Mitigation:* per-configuration wall times and peak memory are stored in the results DBs; peak memory was not measured per stage.

4. **Missing method families.** No learned vectorizers (PersLay, Hofer; #8), no H_2 features (#9), no cross-library replication (#11), no protein/graph modalities. *Severity: moderate* — each is a named boundary of the current claim, and each maps to a registered expansion. The paper's contribution is scoped accordingly: the internal stage-importance decomposition and the framework, not an absolute-accuracy win over classical methods (raw baselines beat TDA on the studied sets; §4.3, §5.2).

Chapter 6

Conclusion and Operational Guidelines

Vectorization is the dominant pipeline stage on both real datasets in this study. On ECG200 time series, vectorization produced a 6.39pp marginal accuracy range across 7 vectorizers (95% CI [6.13, 6.65]), while filtration contributed 0.69pp across 3 filtrations (95% CI [0.57, 0.81]; small but with CI excluding zero under repeated CV) and classifiers 3.50pp (95% CI [3.28, 3.71]); the ordering was stable across all 25 cross-validation repetitions. On binary MNIST the same hierarchy holds — vectorizer range 3.22pp versus filtration 1.65pp, with cubical beating Vietoris-Rips by ∼1.75pp best-of-family — and a second time-series dataset (ECG5000) shows the raw-menu ordering again (24.89pp versus 3.60pp, single split, 3-vectorizer menu); the level-matched comparison reverses there (filtration 3.60pp vs. vectorizer 0.24pp, §4.1). Conti et al.'s image results are a complementary case study on 10-class MNIST under grid search, not a general law. On synthetic sphere/torus data, the combined geometric signal survives $\sigma = 0.30$ additive spatial Gaussian noise at 99.85% mean accuracy across the 112 configurations at that level; measured bottleneck distances (max 0.434 versus the corrected bound $2\sigma\sqrt{2\ln n} \approx 1.82$) confirm the stability bound is conservative, and a matched genus-1/genus-2 experiment with identical norm distributions shows the robustness is not an artefact of the norm confound (TDA 95.83% versus 48–58% for norm features at $\sigma = 0.30$). TDA features do not beat raw features on the datasets studied (raw-signal logistic 85.28% vs. 83.0% TDA on ECG200 under the 25-repetition protocol; raw-pixel logistic 99.65% vs. 98.0% TDA on MNIST), so the contribution is the internal TDA-pipeline comparison.

The core contribution is the conditionality itself: benchmark claims about pipeline-stage importance depend on the datasets, modalities, and fixed dimensions of the study, and those conditions should be stated as explicitly as the claims. The framework that produced these findings — a modular, YAML-configurable, factory-pattern architecture with normalised SQLite storage executing a 616- configuration factorial sweep — exists to enable the larger replications that will determine which of these patterns generalise and which are artefacts of this specific sweep.

6.1 A Decision Tree for Practitioners

The conditionality established in this dissertation collapses into a small decision tree. Each branch names the evidence that supports it and the grade of that evidence: A = verified under the 25-repetition repeated-CV protocol with corrected confidence intervals, or replicated across datasets; B = measured at r=5 or single-split with consistent replication; C = single observation or a sweep that has not yet run. The tree is deliberately conservative — where the evidence is thin, the branch says so rather than pretending otherwise.

1. **Modality.** The first question is the data modality.

   (a) *Time series (after Takens embedding, d=3, τ=1):* spend your design budget on the **vectorizer**, not the filtration. Vectorization is the dominant stage (ECG200 marginal range 6.39pp, 95% CI [6.13, 6.65] over 25 repetitions; ECG5000 24.89pp, single split, 3-vectorizer menu; §4.1, Table 4.2), while filtration contributes 0.69pp (CI [0.57, 0.81]) and classifiers 3.50pp (CI [3.28, 3.71]). *Grade A* for the *ordering*: the vectorizer is the only stage whose CI excludes zero, the ordering holds in 25/25 repetitions, and the ω^2 population effect size (ECG200 vectorizer 0.165, CI [0.063, 0.423], vs. filtration −0.017, Table 4.3) confirms it is not a level-count artefact. *Caveat:* the measured *margin* is menu-sensitive — excluding the scalar vectorizers cuts it to 3.10pp, and matching level counts (best-3 vectorizers vs. 3 filtrations) narrows it to 1.21pp vs. 0.69pp (§4.1). The ordering survives; the size does not.

   (b) *Images, binary:* vectorization still leads (MNIST binary 3.22pp vs. filtration 1.65pp, single split; pooled over the five repetitions 3.03pp vs. 1.55pp), but filtration is no longer negligible — prefer the **cubical** filtration over Vietoris–Rips (best-of-family 98.0% vs. 96.25%; §4.2). *Grade B* (r=5; the vectorizer > filtration ordering holds in all five repetitions).

   (c) *Images, multi-class ($\geq$ 10 classes):* the ordering flips. Filtration becomes the larger marginal stage (10-class MNIST: 4.53pp vs. vectorizer 3.44pp; classifier 2.03pp; cubical 29.1% vs. Vietoris–Rips 33.7% within filtration), with a 9–10pp span within a single filtration — still far below Conti et al.'s 18–94% grid-search swing (§4.2). *Grade B* (r=5, 1000 samples).

   (d) *Point clouds:* on saturated topology tasks no stage matters: at σ=0 all 112 configurations score $\geq$ 99.5% (mean 99.99%) and every stage range is below

0.1pp (vectorizer 0.09, filtration 0.02, classifier 0.05); at σ=0.30 the largest stage range is 0.66pp (vectorizer). *Grade A* for saturation. Where the classes differ in genus or component count, TDA separates them nearly perfectly and the matched-genus control shows the topological signal survives σ=0.30 (TDA 95.83% vs. norm features 48–58%; §4.2). *Grade A*.

2. **Budget: accuracy vs. runtime.** If wall-clock time is the constraint:

   (a) *Large n ($\geq 10^3$ points):* Sparse Rips is the only filtration whose design point applies, and only with an implementation-level benchmark at the target n: at n=1000 it reaches the same 100.00% as Vietoris–Rips (8/8 configurations) at $\approx 30\times$ the cost ($\sim$1.8h vs. $\sim$3.6min), and at n=3000 it did not finish in $\sim$42h on this hardware (§5.3). Vietoris–Rips at n=3000 is likewise infeasible ($O(n^3)$ 2-simplices). *Grade B* (B5 large-n sweep, n=1000: 8/8 finished runs; n=3000: 0/2 finished).

   (b) *Small n:* any of Vietoris–Rips, weak Alpha, or Cubical; the 3–27% wall-clock gap between them (§4.3) rarely justifies accuracy risk. Avoid weak Alpha on quantized UCR series (giotto-tda raises `IndexError`; the fragility is a first-class result of the B2 panel, §5.3). *Grade A* (wall-clock recorded per configuration in the results DBs).

   (c) *Classifier:* prefer Random Forest or logistic regression on TDA features. SVM-RBF collapses to the majority class on ECG200 (66.5%) in both the TDA-only and the concatenated arms (§5.2). *Grade B*.

3. **Noise level.** In the studied regime ($\sigma \leq 0.30$ additive spatial Gaussian noise, n=100), noise robustness is not a differentiator: mean accuracy across the 112 configurations at σ=0.30 is 99.85% (minimum 98.5%), and the measured bottleneck distances (maximum 0.434) sit far below the corrected stability bound ($2\sigma\sqrt{2\ln n} \approx 1.82$) — the bound is conservative, not tight (§4.2). If your application lives in this regime, do not trade accuracy for noise-robust filtrations (DTM-weighting gained menu range but not headroom on ECG200; B1). *Grade A*.

4. **Which vectorizer?** Prefer *distributional* vectorizers over scalar summaries:

   (a) Persistence Entropy — the single-scalar-per-dimension representation — is the weakest vectorizer on ECG200 (68.95% under r=25) and catastrophic on ECG5000 (36.87% vs. 61.5%/ 61.8% for Betti Curve/Silhouette). *Grade A*.

   (b) Top performers by dataset: Persistence Landscapes (ECG200, 75.32%), Betti Curve and Silhouette (ECG5000, 61.5%/61.8%), Betti Curve (10-class

MNIST, 33.4%). *Grade B.*

(c) Persistence Statistics is a zero-parameter, cheap default: within 0.3pp of Landscapes on ECG200 at r=5 (74.8% vs. 75.1%) and within 2.1pp of the best MNIST configuration (95.9% vs. 98.0%; §4.1). *Grade B.*

5. **Raw features vs. TDA.** On the datasets studied, raw features beat TDA alone (ECG200 raw logistic 85.28% and raw RF 86.30% under r=25 vs. 83.0% single-split best TDA, whose repeated-CV mean is 79.6%; MNIST raw-pixel logistic 99.65% vs. 98.0%; §4.3). Do not use TDA as your *only* feature source on such data. If you want the geometric signal, **concatenate**: [raw‖TDA] adds 0.35–1.72pp over the 25-repetition raw baselines (ECG200 87.0% vs. 85.28%; MNIST 100.0% vs. 99.65%), never hurts with logistic or Random Forest, and the benefit is confined to those two classifiers (§5.2). *Grade B* (single protocol, both datasets).

6. **Is topology the target?** If the scientific question is about shape — cycle structure, genus, component counts (dynamical systems, shape/geometry benchmarks) — TDA is not a feature extractor to be benchmarked against raw baselines; it is the object of study. The matched-genus experiment is the proof of concept: TDA retains 95.83% where norm/scale features fail (48–58%). The topology-wins regime (expansion #6) is designed to map this branch with the full stage decomposition; until it runs, treat this branch as *Grade C.*

Table 6.1: Decision tree summary with evidence grades.

Branch	Decision	Evidence (paper)	Grade
Time series	tune vectorizer	ECG200 6.39pp [6.13, 6.65] r=25; ECG5000 24.89pp (§4.1, Table 4.2)	A
Image, binary	vectorizer first; cubical > VR	MNIST 3.22 vs. 1.65pp; 98.0 vs. 96.25% (§4.2)	B
Image, multi-class	filtration first	10-class MNIST 4.53 vs. 3.44pp (§4.2)	B
Point cloud	no stage dominates	all stage ranges < 1pp at $\sigma \leq 0.30$; 99.85% mean at σ=0.30 (§4.2)	A
Speed, large n	Sparse Rips only with impl. benchmark	B5: parity at n=1000 at $\sim$30$\times$ cost; n=3000 infeasible (§5.3)	B
Speed, small n	VR/weak Alpha/Cubical; no weak Alpha on quantized UCR	wall-clock in DBs (§4.3); fragility finding (B2, §5.3)	A
Classifier	RF / logistic; not SVM-RBF on TDA features	66.5% majority-class collapse, both arms (§5.2)	B
Noise ≤ 0.30	no robustness trade-off needed	99.85% mean; bottleneck $0.434 \ll 1.82$ (§4.2)	A
Vectorizer	distributional > scalar; entropy last	entropy 68.95%/36.87%; landscape 75.32% (Tables 4.2, §4.1–§4.2)	A
Raw available	concat [raw‖TDA] with RF/logistic	+0.35–1.72pp over 25-rep raw baselines (§5.2)	B
Topology is the target	TDA as object of study	matched-genus 95.83 vs. 48–58% (§4.2); #6 pending	C

Appendix A

Full YAML Configuration

```yaml
# Expanded benchmark -- covers full framework surface area.
# Non-image filtrations silently fail on image data and vice versa;
# the runner catches and logs failures per-config.
datasets:
  - name: sphere_torus_n0
    path: data/tda/synthetic/sphere_torus_noise0_X.npy
    labels: data/tda/synthetic/sphere_torus_noise0_y.npy
    modality: point_cloud
    max_samples: 200
    subsample_points: 100
  - name: sphere_torus_n5
    path: data/tda/synthetic/sphere_torus_noise5_X.npy
    labels: data/tda/synthetic/sphere_torus_noise5_y.npy
    modality: point_cloud
    max_samples: 200
    subsample_points: 100
  - name: sphere_torus_n15
    path: data/tda/synthetic/sphere_torus_noise15_X.npy
    labels: data/tda/synthetic/sphere_torus_noise15_y.npy
    modality: point_cloud
    max_samples: 200
    subsample_points: 100
  - name: sphere_torus_n30
    path: data/tda/synthetic/sphere_torus_noise30_X.npy
    labels: data/tda/synthetic/sphere_torus_noise30_y.npy
    modality: point_cloud
    max_samples: 200
    subsample_points: 100
  - name: ecg200
    path: data/tda/ucr/ecg200_X.npy
    labels: data/tda/ucr/ecg200_y.npy
    modality: time_series
    takens_dimension: 3
```

```yaml
    takens_delay: 1
  - name: mnist_01
    path: data/tda/images/mnist_01_X.npy
    labels: data/tda/images/mnist_01_y.npy
    modality: image
    max_samples: 400
    description: "MNIST binary (0 vs 1) -- 200 per class, 28x28 greyscale"

filtrations:
  - name: vietoris_rips
    homology_dimensions: [0, 1]
  - name: weak_alpha
    homology_dimensions: [0, 1]
  - name: sparse_rips
    homology_dimensions: [0, 1]
    epsilon: 0.3
  - name: cubical
    homology_dimensions: [0, 1]

vectorizations:
  - name: persistence_image
    sigma: 0.1
    n_bins: 20
  - name: persistence_landscape
    n_layers: 3
    n_bins: 50
  - name: betti_curve
    n_bins: 50
  - name: silhouette
    n_bins: 50
  - name: persistence_entropy
    normalize: true
  - name: amplitude
    metric: bottleneck
  - name: persistence_statistics

classifiers:
  - name: svm_rbf
  - name: random_forest
  - name: logistic
  - name: svm_linear

evaluation:
  cv_folds: 5
  scoring: accuracy
  random_seed: 42
```

```
  repetitions: 1

output:
  db_path: data/tda/expanded_results.db
  log_level: WARNING
```

Appendix B

SQLite Schema Reference

```sql
CREATE TABLE IF NOT EXISTS runs (
    run_id      INTEGER PRIMARY KEY AUTOINCREMENT,
    dataset     TEXT    NOT NULL,
    filtration  TEXT    NOT NULL,
    vectorizer  TEXT    NOT NULL,
    classifier  TEXT    NOT NULL,
    repetition  INTEGER NOT NULL,
    started_at  TEXT,
    finished_at TEXT,
    wall_time_s REAL,
    peak_memory_mb REAL
);

CREATE TABLE IF NOT EXISTS fold_results (
    fold_id     INTEGER PRIMARY KEY AUTOINCREMENT,
    run_id      INTEGER NOT NULL REFERENCES runs(run_id),
    fold        INTEGER NOT NULL,
    accuracy    REAL,
    f1          REAL,
    precision   REAL,
    recall      REAL
);

CREATE TABLE IF NOT EXISTS run_metadata (
    run_id          INTEGER PRIMARY KEY REFERENCES runs(run_id),
    pipeline_params TEXT,
    n_train         INTEGER,
```

```sql
    n_test          INTEGER,
    n_features      INTEGER
);

CREATE TABLE IF NOT EXISTS config_snapshot (
    id          INTEGER PRIMARY KEY CHECK (id = 1),
    yaml_text   TEXT    NOT NULL,
    saved_at    TEXT    NOT NULL
);

CREATE INDEX IF NOT EXISTS idx_runs_dataset ON runs(dataset);
CREATE INDEX IF NOT EXISTS idx_runs_filtration ON runs(filtration);
CREATE INDEX IF NOT EXISTS idx_fold_run ON fold_results(run_id);
```

Appendix C

Code Implementation

The factory classes and runner are available at `scripts/tda_benchmark/`. Key excerpts follow.

FiltrationFactory

```
class FiltrationFactory:
    @staticmethod
    def create(name: str, **kwargs):
        mapping = {
            "vietoris_rips": VietorisRipsPersistence,
            "sparse_rips": SparseRipsPersistence,
            "cubical": CubicalPersistence,
            "weak_alpha": WeakAlphaPersistence,
        }
        cls = mapping.get(name)
        if cls is None:
            raise ValueError(f"Unknown␣filtration:␣{name}")
        return cls(n_jobs=1, **kwargs)
```

Listing C.1: FiltrationFactory.

VectorizationFactory (auto-flatten)

```
if name != "persistence_statistics":
    return Pipeline([
        ("vec", vec),
        ("flatten", FunctionTransformer(
            lambda x: x.reshape(x.shape[0], -1),
            validate=False)),
    ])
return vec
```

Listing C.2: VectorizationFactory auto-flatten.

Execution Loop

```
for ds, fil, vec, clf in product(datasets, filtrations,
                                  vectorizers, classifiers):
    pipeline = Pipeline([
        ("filtration",  FiltrationFactory.create(fil.name, ...)),
        ("vectorizer",  VectorizationFactory.create(vec.name, ...)),
        ("classifier",  ClassifierFactory.create(clf.name, ...)),
    ])
    scores = cross_validate(pipeline, X, y,
        cv=StratifiedKFold(n_splits=cv_folds, shuffle=True,
                     random_state=random_seed + rep)        scoring=["
                         accuracy", "f1_weighted",
                     "precision_weighted", "recall_weighted"])
```

Listing C.3: PipelineRunner execution loop.

PersistenceStatistics

```
class PersistenceStatistics(BaseEstimator, TransformerMixin):
    def transform(self, X):
        features = []
        for d in sorted(dims):
            mask = X[:, :, 2] == d
            births, deaths = X[:,:,0][mask], X[:,:,1][mask]
            lifespans = deaths - births
            for arr in [births, deaths, lifespans]:
                features.extend([np.nanmean(arr), np.nanstd(arr),
                                 np.nanmin(arr), np.nanmax(arr)])
        return np.column_stack(features)
```

Listing C.4: PersistenceStatistics transformer.

Appendix D

Extended Parameter Sweeps

The following sensitivity tests were conducted on reduced configuration subsets.

> **Provenance note.** The Takens sensitivity sweep below is produced by the repository's own runner worker on the full grid $d \in \{2, 3, 4\} \times \tau \in \{1, 2, 3\}$ (72 grid cells; `data/tda/takens_sweep.db`, CV seeds 43), additive and fully reproducible. At d=2 the weak-alpha filtration on 2-D embedded clouds produces essential H_1 classes (death = $+\infty$), so the six d=2 weak-alpha × persistence-image cells crash giotto-tda natively (the infinite deaths reach PersistenceImage's internal integer cast). These cells were completed by clipping the infinite deaths to the maximum finite death (flagged in the database, runs 73–78), and the table below includes these clipped cells. This clipping is the standard treatment equivalent to giotto's own landscape/Betti handling of essential classes.

Takens Embedding Sensitivity

The stage hierarchy of §4.1 was tested under alternative embedding parameters on a reduced pipeline subset (2 filtrations × 2 vectorizers × 2 classifiers):

Table D.1: Stage marginal ranges under alternative (d, τ) (reduced 8-configuration subset; range of stage-level means).

(d, τ)	Fil. Range	Vec. Range	Clf. Range
(2, 1)	0.38pp	2.12pp	7.62pp
(2, 2)	1.00pp	1.75pp	7.75pp
(2, 3)	0.12pp	2.62pp	3.88pp
(3, 1)	1.12pp	1.38pp	9.62pp
(3, 2)	1.00pp	6.00pp	8.75pp
(3, 3)	0.12pp	2.88pp	7.38pp
(4, 1)	0.50pp	5.50pp	6.50pp
(4, 2)	0.00pp	3.75pp	8.25pp
(4, 3)	1.00pp	2.00pp	7.25pp

On this reduced subset the classifier is the largest stage (3.88–9.63pp; driven by SVM-RBF collapsing to the 66.5% majority class on Persistence Entropy features) and the vectorizer second (1.38–6.00pp), while the filtration range stays at or below 1.13pp for every (d, τ) pair. The filtration-invariance of the stage analysis therefore holds under alternative embeddings; the vectorization-dominance ordering of §4.1 is measured on the full 112-configuration sweep and does not transfer to this deliberately reduced 8-configuration subset, where the vectorizer/classifier interaction dominates.

Framework Extensibility

The framework supports 7 filtrations (VR, Alpha, Sparse Rips, Čech, Cubical, Weighted Rips, Flagser), 11 vectorizers (PI, PL, Betti, Silhouette, Entropy, Amplitude, Heat Kernel, Complex Polynomials, Pairwise Distance, Number of Points, Persistence Statistics), and 4 classifiers (SVM-linear, SVM-RBF, RF, Logistic). The 616-configuration sweep reported in Chapter 4 exercises 4, 7, and 4 of these respectively. Adding a new method to the sweep requires adding one entry to the YAML configuration (Appendix A). Adding a fundamentally new component type (e.g., a learned vectorizer) requires a new factory entry in `factories.py`.

Appendix E

Use of AI Tools

AI tools were used in the following capacities:

- **Code generation and debugging.** An AI coding assistant assisted with implementing the benchmarking framework: factory-pattern architecture, YAML loader, SQLite storage, pipeline runner, and analysis module. All code was reviewed, tested, and modified by me.
- **Literature synthesis.** AI tools assisted in extracting key findings from the prior-art survey.
- **Document preparation.** The structural template was adapted from previous work with AI assistance.

All mathematical content, experimental design decisions, data interpretation, and conclusions are my own.

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